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INFORMATION PROCESSING APPARATUS, INFORMATION PROCESSING METHOD, AND FIRMNESS ALTERABLE MODULE

Abstract

An information processing apparatus according to the present disclosure includes: an acquisition unit that acquires sensor information detected by a sensor; an estimation unit that estimates, based on the sensor information acquired by the acquisition unit, a contact mode of an object with respect to a target of which a firmness of a surface is alterable; and a determination unit that determines whether to alter the firmness of the surface of the target according to the contact mode of the object estimated by the estimation unit.

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Background/Summary

FIELD

[0001] The present disclosure relates to an information processing apparatus, an information processing method, and a firmness alterable module.

BACKGROUND

[0002] There is a conventionally known technique for altering a state of an object according to user's preference. For example, there is provided a technology that enables a user of a bed, which is an example of the object, to select a bed having firmness suitable for the user's own preference.

CITATION LIST

Patent Literature

[0003] Patent Literature 1: JP 2000-106979 A

SUMMARY

Technical Problem

[0004] However, the known technology cannot always ensure appropriate alteration of the firmness of the surface of the object. For example, the known technology needs an explicit instruction by the user in selecting the firmness, and the user needs to explicitly select the firmness each time the firmness is altered, and there is room for improvement in terms of usability. In addition, the known technology has not considered an issue of the shape of the surface of the object affected by adjusting the firmness. In this regard, the shape of the surface of the object greatly changes depending on the firmness, making it difficult to maintain a desired shape of the object surface. In this manner, when the firmness of the surface of an object is altered, there is room for improvement in terms of usability and the shape of the surface, and it is desired to be able to appropriately alter the firmness of the surface of an object.

[0005] In view of this, the present disclosure proposes an information processing apparatus, an information processing method, and a firmness alterable module capable of appropriately altering the firmness of the surface of an object.

Solution to Problem

[0006] According to the present disclosure, an information processing apparatus includes an acquisition unit that acquires sensor information detected by a sensor; an estimation unit that estimates, based on the sensor information acquired by the acquisition unit, a contact mode of an object with respect to a target of which a firmness of a surface is alterable; and a determination unit that determines whether to alter the firmness of the surface of the target according to the contact mode of the object estimated by the estimation unit.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] FIG. 1 is a diagram illustrating an example of information processing according to an embodiment of the present disclosure.

[0008] FIG. 2 is a diagram illustrating a configuration example of an information processing system according to the embodiment of the present disclosure.

[0009] FIG. 3 is a diagram illustrating a configuration example of an information processing apparatus according to the embodiment of the present disclosure.

[0010] FIG. 4 is a diagram illustrating an example of a module information storage unit according to the embodiment of the present disclosure.

[0011] FIG. 5 is a diagram illustrating an example of a history information storage unit according to the embodiment of the present disclosure.

[0012] FIG. 6 is a diagram illustrating an example of a sensor information storage unit according to the embodiment of the present disclosure.

[0013] FIG. 7 is a diagram illustrating an example of an alteration condition information storage unit according to the embodiment of the present disclosure.

[0014] FIG. 8 is a diagram illustrating a configuration example of a firmness alterable module according to the embodiment of the present disclosure.

[0015] FIG. 9 is a plan view illustrating a main part of the firmness alterable module.

[0016] FIG. 10 is a side view illustrating the main part of the firmness alterable module.

[0017] FIG. 11 is a side view illustrating the main part of the firmness alterable module.

[0018] FIG. 12 is a diagram illustrating an arrangement example of a plurality of firmness alterable modules.

[0019] FIG. 13 is a flowchart illustrating a procedure of information processing according to the embodiment of the present disclosure.

[0020] FIG. 14 is a flowchart illustrating a procedure of processing according to pressure.

[0021] FIG. 15 is a diagram illustrating an example of alteration according to an adjacent module.

[0022] FIG. 16 is a flowchart illustrating a procedure of processing according to a pressure change pattern.

[0023] FIG. 17 is a conceptual diagram illustrating an example of alteration according to object detection.

[0024] FIG. 18 is a flowchart illustrating a procedure of processing according to object detection.

[0025] FIG. 19 is a conceptual diagram illustrating an example of alteration according to a context.

[0026] FIG. 20 is a conceptual diagram illustrating an example of alteration according to a context.

[0027] FIG. 21 is a flowchart illustrating a procedure of processing according to a context.

[0028] FIG. 22 is a flowchart illustrating a procedure of processing according to a user.

[0029] FIG. 23 is a conceptual diagram illustrating an example of alteration according to a physical part of a user.

[0030] FIG. 24 is a flowchart illustrating a procedure of processing according to a physical part of a user.

[0031] FIG. 25 is a conceptual diagram illustrating an example of alteration according to a force applied to a user.

[0032] FIG. 26 is a flowchart illustrating a procedure of processing according to a force applied to a user.

[0033] FIG. 27 is a hardware configuration diagram illustrating an example of a computer that implements an information processing apparatus or functions of the information processing apparatus.

DESCRIPTION OF EMBODIMENTS

[0034] Embodiments of the present disclosure will be described below in detail with reference to the drawings. Note that the information processing apparatus, the information processing method, and the firmness alterable module according to the present application are not limited by the embodiments. Moreover, in each of the following embodiments, the same parts are denoted by the same reference symbols, and a repetitive description thereof will be omitted.

[0035] The present disclosure will be described in the following order. [0036] 1. Embodiments

[0037] 1-1. Overview of information processing according to embodiment of present disclosure

[0038] 1-2. Configuration of information processing system according to embodiment [0039] 1-3.

Configuration of information processing apparatus according to embodiment [0040] 1-4.

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example of object using firmness alterable module [0042] 1-5. Processing example [0043] 1-5-1. Information processing procedure according to embodiment [0044] 1-5-2. Processing example according to pressure [0045] 1-5-3. Example of processing according to object detection [0046] 1-5-4. Processing example according to context [0047] 1-5-5. Processing example according to user [0048] 1-5-6. Processing example according to force applied to user [0049] 1-5-7. Other information examples [0050] 2. Other embodiments [0051] 2-1. Other configuration examples 2-2. Others [0052] 3. Effects according to present disclosure [0053] 4. Hardware configuration

1. EMBODIMENTS

1-1. Overview of Information Processing According to Embodiment of Present Disclosure [0054] FIG. 1 is a diagram illustrating an example of information processing according to an embodiment of the present disclosure. The information processing according to the embodiment of the present disclosure is implemented by an information processing system 1 including an information processing apparatus 100 and a firmness alterable module 10 (hereinafter, also simply referred to as a “module”) as illustrated in FIG. 2. Details of the information processing system 1 will be described with reference to FIG. 2.

[0055] FIG. 1 will be used to describe an exemplary case where an object 2, which is a target in which the firmness of the surface is alterable, is a sofa. However, the object 2 is not limited to a sofa, and may be any object as long as the object 2 is a tangible object having a face (surface) with which another object comes into contact. For example, the object 2 may be a desk, an automobile seat, a cushion, or the like. In FIG. 1, the object 2 before the firmness alteration is described as an “object 2-1”, and the object 2 after the firmness alteration is described as an “object 2-2”. However, the object 2 will be described as the object 2 when described with no particular distinction from each other. The object 2-1 on which the user U1 sits in FIG. 1 indicates the object 2 as a side view, and other objects 2-1 and 2-2 indicate plan views (top views) of the object 2 viewed from above the seat surface of the object 2.

[0056] In FIG. 1, the object 2 being a sofa has its seat surface on which a person sits includes a plurality of firmness alterable modules 10. Specifically, the seat surface of the object 2 includes 35 firmness alterable modules 10 indicated by 5×7 rectangular regions. Incidentally, the object 2 being a sofa may have the seat surface on which a person sits constituted with one firmness alterable module 10.

[0057] FIG. 1 assigns M and two-digit numbers such as “M **” (* is a number) to 5×7 rectangular regions each corresponding to each of 35 firmness alterable modules 10 so as to make each of the firmness alterable modules 10 distinguishable from each other. Specifically, in FIG. 1, M11 to M57 are assigned individually to the 5×7 rectangular regions so as to distinguishably describe the 35 firmness alterable modules 10 (hereinafter also referred to as “modules M11 to M57”) provided in the object 2 being a sofa.

[0058] For example, the module M11 is the firmness alterable module 10 located at the left corner of the object 2 in plan view from the seat surface side of the object 2. The module M21 is the firmness alterable module 10 disposed on the upper side of the module M11 in plan view from the seat surface side of the object 2. The module M12 is the firmness alterable module 10 disposed on the right side of the module M11 in plan view from the seat surface side of the object 2.

[0059] For example, the module M34 is the firmness alterable module 10 located at the center of the object 2 in plan view from the seat surface side of the object 2. The module M24 is the firmness alterable module 10 disposed on the lower side of the module M34 in plan view from the seat surface side of the object 2. The module M33 is the firmness alterable module 10 disposed on the left side of the module M34 in plan view from the seat surface side of the object 2.

[0060] For example, the module M11 corresponds to a firmness alterable module 10-11 in FIG. 2, and the module M12 corresponds to a firmness alterable module 10-12 in FIG. 2. In addition, the module M13 corresponds to a firmness alterable module 10-13 in FIG. 2, the module M14 corresponds to a firmness alterable module 10-14 in FIG. 2, and the module M15 corresponds to a

firmness alterable module **10-15** in FIG. 2. Each of the modules **M11** to **M57** is described as the firmness alterable module **10** or simply the module when described with no particular distinction from each other.

[0061] First, when the user **U** sits on the seat surface of the object **2** as a sofa, a pressure applied to the seat surface of the object **2** is detected. FIG. 1 illustrates, as an example, a case where a user **U** sits on regions corresponding to modules **M13**, **M14**, **M15**, **M23**, **M24**, and **M25** on the seat surface of an object **2**. For example, the object **2** is provided with a pressure sensor corresponding to each of the 35 modules, and each pressure sensor detects the pressure applied to each of the 35 modules provided on the seat surface of the object **2**. Each of the modules includes a pressure sensor and transmits the detected pressure to the information processing apparatus **100**. With this configuration, the information processing apparatus **100** acquires sensor information indicating the pressure applied to the object **2** by the user **U** (step **S1**).

[0062] The information processing apparatus **100** determines whether to alter the firmness of the surface of the object **2** based on the acquired sensor information (step **S2**). In FIG. 1, the information processing apparatus **100** determines which region of the seat surface of the object **2** is to undergo firmness alteration based on the acquired sensor information. The information processing apparatus **100** estimates a contact mode of a user **U1** with respect to the object **2**. The information processing apparatus **100** calculates a change amount in the pressure applied by the user **U** to each of the 35 modules of the object **2** based on the change amount in the pressure detected by each pressure sensor.

[0063] The information processing apparatus **100** determines a module in which the change amount in the pressure applied by the user **U** satisfies a condition, among the 35 modules of the object **2**, as a module to undergo firmness alteration. For example, the information processing apparatus **100** determines a module in which the change amount in the pressure applied by the user **U** is a predetermined threshold or more, among the 35 modules of the object **2**, as a module to undergo firmness alteration (also referred to as “alteration target module”).

[0064] In FIG. 1, the information processing apparatus **100** determines six modules of modules **M13**, **M14**, **M15**, **M23**, **M24**, and **M25** as alteration target modules among 35 modules provided in the object **2** being a sofa. That is, the information processing apparatus **100** determines to soften, that is, decrease the firmness of the six modules **M13**, **M14**, **M15**, **M23**, **M24**, and **M25** corresponding to the regions of the object **2** where the user **U** sits.

[0065] Furthermore, the information processing apparatus **100** may determine that the larger the change amount in the pressure of the alteration target module, the larger the alteration amount in the firmness of the alteration target module. In FIG. 1, the information processing apparatus **100** may determine the alteration amount in the firmness of the module **M24** so that the module **M24** is to be softer than others among the six modules of the modules **M13**, **M14**, **M15**, **M23**, **M24**, and **M25** which are alteration target modules.

[0066] The information processing apparatus **100** transmits instruction information (control information) instructing firmness alteration to each alteration target module. The information processing apparatus **100** transmits instruction information indicating a firmness alteration amount to each alteration target module. In FIG. 1, the information processing apparatus **100** transmits instruction information instructing the firmness alteration to six modules of modules **M13**, **M14**, **M15**, **M23**, **M24**, and **M25**, which are alteration target modules. For example, the information processing apparatus **100** transmits, to the module **M13**, instruction information instructing firmness alteration on the surface of the module **M13**.

[0067] Having received the instruction information from the information processing apparatus **100**, each alteration target module alters the firmness of the surface based on the instruction information. The six modules of the modules **M13**, **M14**, **M15**, **M23**, **M24**, and **M25** alter the firmness of the surface based on the instruction information. For example, the module **M13** executes processing of altering the firmness of the surface based on the instruction information received from the

information processing apparatus **100**. With this configuration, the object **2** can soften regions corresponding to the six modules of the modules **M13**, **M14**, **M15**, **M23**, **M24**, and **M25** in the seat surface. Details of the configuration of implementing the alteration of firmness in the firmness alterable module **10** will be described below.

[0068] As described above, based on the sensor information, the information processing apparatus **100** determines whether to alter the firmness of the surface of the object **2** whose firmness of the surface is alterable, making it possible to appropriately alter the firmness of the surface of the object. Furthermore, the firmness alterable module **10** provided in the object **2** alters the firmness in accordance with an instruction from the information processing apparatus **100**, making it possible to appropriately alter the firmness of the surface of the object.

[0069] Conventionally, the firmness of a seat surface of a product such as desk or a sofa is a firmness unique to the product. However, depending on the use condition, there are cases where the firmer one is more comfortable to use and where the softer one is better. For example, in the case of a desk, it is more appropriate to have a firm flat surface as usual when writing on the desk, but it is more appropriate to have a soft surface such as a cushion when sleeping with one's head down or placing an object that easily rolls.

[0070] On the other hand, it is possible to increase the firmness of a soft material by compressing the material, but it is difficult to maintain the surface shape. Moreover, it is possible to solve the issue of maintaining the surface shape by stacking a material different from the compressive material on the surface, but it is difficult to express softness with a material having firmness enough to maintain the surface shape.

[0071] In view of these, it is desirable to have a mechanism that can withstand deformation during compression and behaves to provide softness in normal times. Therefore, the information processing system **1** uses a structure in which softness is achieved not by firmness of the material but by a structure having a desired behavior so as to be able to withstand deformation during compression and behaves to provide softness in normal times. In the information processing system **1**, the structure of the firmness alterable module **10** is a structure that exhibits behavior that is deformed by a force from the positive direction of the axis and is unlikely to be deformed by a force from the negative direction. With this structure, the information processing system **1** can actualize an interface that can adaptively alter the firmness using a mechanism that can alter the firmness while maintaining the surface shape. The information processing system **1** can perform driving at a deformation speed at which firmness can be altered while maintaining a surface shape and which can cope with daily movements of a person.

[0072] As described above, by using the firmness alterable module **10** having a structure that exhibits a behavior unlikely to be deformed against a force applied during compression, the information processing system **1** can actualize a mechanism capable of maintaining a surface shape during compression while maintaining softness during normal times. Details of the structure of the firmness alterable module **10** will be described below.

1-2. Configuration of Information Processing System According to Embodiment

[0073] The information processing system **1** illustrated in FIG. **2** will be described. FIG. **2** is a diagram illustrating a configuration example of the information processing system according to the embodiment of the present disclosure. The information processing system **1** illustrated in FIG. **2** may include a plurality of information processing apparatuses **100** and a plurality of sensor apparatuses **50**. As illustrated in FIG. **2**, the information processing system **1** includes the information processing apparatus **100**, a plurality of firmness alterable modules **10-11**, **10-12**, **10-13**, **10-14**, and **10-15**, and the sensor apparatus **50**. Hereinafter, when the firmness alterable modules **10-11** to **10-15** may be referred to as the firmness alterable module **10** when not distinguished from each other.

[0074] FIG. **2** illustrates a case where the information processing apparatus **100** is disposed in the object **2**. Although FIG. **2** illustrates only the five firmness alterable modules **10-11** to **10-15**, the

object **2** may include more than five (for example, 20 or 100 or more) firmness alterable modules **10**. Furthermore, the object **2** may include four or less firmness alterable modules **10**. For example, the object **2** may include one firmness alterable module **10**.

[0075] The information processing apparatus **100** controls the firmness alterable module **10**. For example, the information processing apparatus **100** is connected to the firmness alterable module **10** by wired connection, and transmits and receives information to and from the firmness alterable module **10**. Furthermore, the information processing apparatus **100** is communicably connected to the sensor apparatus **50** by a wired or wireless connection via a predetermined network N.

[0076] The information processing apparatus **100** is a computer that determines whether to alter the firmness of the surface of the target in which the firmness of the surface is alterable, based on sensor information detected by a sensor. The information processing apparatus **100** is used to provide a service related to alteration in firmness of the object **2** equipped with the firmness alterable module **10**. The information processing apparatus **100** performs various types of information processing related to the alteration in firmness of the firmness alterable module **10**.

[0077] The firmness alterable module **10** includes the surface constituting at least a part of the surface of the object **2**. In FIG. **1**, the firmness alterable module **10** constitutes a seat surface portion of the object **2** being a sofa. The firmness alterable module **10** is a module capable of altering its firmness. Details of the configuration of the firmness alterable module **10** will be described with reference to FIG. **8** and subsequent drawings.

[0078] The sensor apparatus **50** detects various types of sensor information. The sensor apparatus **50** may be an image sensor (camera) that captures an image. The sensor apparatus **50** being an image sensor is disposed around the object **2** and images the object **2**. For example, the sensor apparatus **50** being an image sensor images another object such as a person in contact with the object **2**.

[0079] Note that the sensor apparatus **50** is not limited to the image sensor, and may be various sensors. For example, the sensor apparatus **50** may be a sound sensor (microphone) that detects sound. For example, the sensor apparatus **50** being a sound sensor collects user's utterance, ambient environmental sound, and the like.

[0080] Furthermore, the sensor apparatus **50** may be a sensor that detects various biological signals, such as a heart rate sensor that detects a heart rate of the user, an electroencephalogram sensor that detects an electroencephalogram of the user, a pulse sensor (pulse wave sensor) that detects a pulse of the user, a respiration sensor (exhalation sensor) that detects respiration of the user, and a perspiration sensor that detects perspiration of the user. Furthermore, the sensor apparatus **50** may be a sensor that detects various types of information such as temperature, humidity, illuminance, position, acceleration, light, pressure, gyro, and distance.

[0081] The sensor apparatus **50** transmits various sensor information detected by various sensors to the information processing apparatus **100**. Note that the above is an example, and the sensor apparatus **50** is not limited to the above, and may be various sensors. In addition, the sensor to detect the various types of information described above in the sensor apparatus **50** may be the same type of sensors or may be different types of sensors.

1-3. Configuration of Information Processing Apparatus According to Embodiment

[0082] Next, a configuration of an information processing apparatus **100** which is an example of an information processing apparatus that executes information processing according to an embodiment will be described. FIG. **3** is a diagram illustrating a configuration example of the information processing apparatus **100** according to the embodiment of the present disclosure.

[0083] As illustrated in FIG. **3**, the information processing apparatus **100** includes a communication unit **110**, a storage unit **120**, and a control unit **130**. The information processing apparatus **100** may include an input unit (for example, a keyboard, a mouse, etc.) that receives various operations from an administrator or the like of the information processing apparatus **100**, and a display unit (for example, a liquid crystal display, etc.) for displaying various types of information.

[0084] The communication unit **110** is implemented by a network interface card (NIC), for example. The communication unit **110** is connected to the network N (refer to FIG. 2) in a wired or wireless channel, and transmits/receives information to/from another information processing apparatus such as the sensor apparatus **50**. The communication unit **110** may communicate with the firmness alterable module **10** and transmit and receive information to and from the firmness alterable module **10**.

[0085] The storage unit **120** is implemented by semiconductor memory elements such as random access memory (RAM) and flash memory, or other storage devices such as a hard disk or an optical disc. As illustrated in FIG. 3, the storage unit **120** according to the embodiment includes a module information storage unit **121**, a history information storage unit **122**, a sensor information storage unit **123**, an alteration condition information storage unit **124**, and a user information storage unit **125**.

[0086] The module information storage unit **121** according to the embodiment stores various types of information related to the module. For example, the module information storage unit **121** can communicate with the information processing apparatus **100** and stores various types of information of modules that can be target modules. FIG. 4 is a diagram illustrating an example of a module information storage unit according to the embodiment of the present disclosure. The module information storage unit **121** illustrated in FIG. 4 includes items such as “module ID”, “placement location”, and “related information”.

[0087] The “module ID” indicates identification information for identifying a module. The “module ID” is provided in the object and indicates identification information for identifying each module to be controlled by the information processing apparatus **100**. Further, the “placement location” indicates an arrangement place of the module. “Placement location” indicates a placement location of the module in the object and a location at which the surface is to be formed by the module.

[0088] The “related information” stores information related to the module. For example, the “related information” stores various types of information indicating a relationship with another module, a sensor that detects the module, and the like. The “related information” includes items such as “adjacent module” and “corresponding sensor”.

[0089] The “adjacent module” indicates a module adjacent to the module (adjacent module). For example, the “adjacent module” stores information (module ID) identifying an adjacent module which is adjacent to the module on its side surface. For example, “adjacent module” indicates modules disposed in four directions, namely, upper, lower, left, and right directions of the module in plan view.

[0090] “Corresponding sensor” indicates a sensor (corresponding sensor) that performs detection related to the module. The “corresponding sensor” stores information (sensor ID or the like) identifying a corresponding sensor. For example, the “corresponding sensor” stores information identifying a pressure sensor that detects pressure on the module.

[0091] FIG. 4 illustrates that the module (module **M11**) identified by the module ID “**M11**” is disposed at a placement location **AP11**. For example, the placement location **AP11** indicates that the module **M11** is placed at the lower left corner of the object **2**. For example, the module **M11** corresponds to the firmness alterable module **10-11**.

[0092] In addition, the diagram indicates that the adjacent module of the module **M11** is a module (module **M12**) identified by the module ID “**M12**” and a module (module **M21**) identified by the module ID “**M21**”. The corresponding sensor of the module **M11** is indicated as a sensor identified by the sensor ID “**SN11**”. For example, the diagram indicates that the corresponding sensor of the module **M11** is a pressure sensor identified by the sensor ID “**SN11**” and that the pressure sensor identified by the sensor ID “**SN11**” detects the pressure on the module **M11**. That is, the pressure value detected by the pressure sensor identified by the sensor ID “**SN11**” indicates the pressure applied to the module **M11**.

[0093] The module information storage unit **121** may store various types of information according to a purpose, not limited to the above.

[0094] The history information storage unit **122** according to the embodiment stores various types of information related to a use history of the object. For example, the history information storage unit **122** stores history information including information such as a user who has used the object, a use date and time, and detailed information regarding the alteration. FIG. 5 is a diagram illustrating an example of a history information storage unit according to the embodiment of the present disclosure. The history information storage unit **122** illustrated in FIG. 5 includes items such as “history ID”, “user”, “date and time”, and “alteration information”.

[0095] “History ID” indicates identification information for identifying history information. The “user” indicates identification information for identifying an entity using an object. When the firmness is altered in the absence of user and there is no corresponding user, the “user” stores information (NULL or the like) indicating that there is no corresponding user. The “date and time” indicates a date and time corresponding to each history ID. For example, “date and time” indicates a date and time of occurrence of alteration corresponding to each history ID. In FIG. 5, “date and time” is illustrated in an abstract manner such as “DA1-1”, but a specific date and time such as “17:36:52, Apr. 11, 2022” may be stored.

[0096] The “alteration details” stores alteration information indicating details of the alteration in firmness of an object. The “history information” includes items such as a “target module”, a “contact mode”, and a “alteration amount”. The “target module” indicates a module being a target of the firmness alteration. A “contact mode” refers to a mode of contact made by another object with the corresponding module. For example, the “contact mode” stores information related to contact made by another objects with the corresponding module, such as a change in pressure to the module. The “alteration amount” indicates alteration amount in the firmness of the corresponding module. For example, the “alteration amount” stores information indicating to which firmness the module is to be altered.

[0097] In FIG. 5, the history information (history information LG1-1) identified by the history ID “LG1-1” indicates alteration in firmness performed on the date and time DA1-1 by the user identified as the user U1. the alteration in the firmness corresponding to the history information LG1-1 indicates that the target module is the module M11. The history information LG1-1 indicates that the firmness of the module M11 having a contact mode CV1 at the date and time DA1-1 has been altered to the firmness corresponding to the alteration amount MV1.

[0098] Note that the history information storage unit **122** may store various types of information according to a purpose, not limited to the above.

[0099] The sensor information storage unit **123** according to the embodiment stores various types of information related to the sensor. FIG. 6 is a diagram illustrating an example of a sensor information storage unit according to the embodiment of the present disclosure. For example, the sensor information storage unit **123** stores various sensor information detected by the sensor apparatus **50**. The sensor information storage unit **123** illustrated in FIG. 6 includes items such as “detection ID”, “date and time”, and “sensor information”.

[0100] The “detection ID” indicates identification information for identifying the acquired sensor information. The “date and time” indicates a date and time corresponding to each detection ID. For example, the “date and time” indicates date and time of acquisition of the sensor information corresponding to each detection ID. In the example of FIG. 6, the “date and time” is illustrated in an abstract manner such as “DA11-1”, but a specific date and time such as “16:47:28, Apr. 11, 2022” may be stored.

[0101] The “sensor information” indicates detected sensor information. The “sensor information” stores various types of information according to the type of detected information. For example, the “sensor information” stores image information, pressure information indicating pressure, and the like.

[0102] In the example of FIG. 6, the detection identified by the detection ID “DL11-1” (detection DL11-1) indicates sensing corresponding to the date and time DA11-1. The detection DL11-1 indicates that the sensor information SD1-1 is acquired (detected).

[0103] The sensor information storage unit 123 may store various types of information according to a purpose, not limited to the above. For example, the sensor information storage unit 123 may store information identifying the sensor apparatus 50 corresponding to each detection. For example, the sensor information storage unit 123 may store information indicating the position of the sensor apparatus 50 at the date and time corresponding to each detection ID. For example, the sensor information storage unit 123 may store position information such as latitude and longitude of the sensor apparatus 50 at a date and time corresponding to each detection ID.

[0104] The alteration condition information storage unit 124 according to the embodiment stores various types of information related to the firmness alteration condition. The alteration condition information storage unit 124 stores various types of information related to a condition used for judging whether to alter the firmness. FIG. 7 is a diagram illustrating an example of an alteration condition information storage unit according to the embodiment. The alteration condition information storage unit 124 illustrated in FIG. 7 includes items such as “condition ID”, “application”, “condition details”, and “alteration details”.

[0105] The “condition ID” indicates identification information for identifying the condition. “Application” indicates application of the condition. The “condition details” stores information used for judgment (determination) as to whether to alter the firmness. FIG. 7 illustrates an example of storing conceptual information such as “CINF1” in the “condition details”. However, information actually stored includes a threshold used for judgment, information indicating a specific condition such as when to alter firmness, or a file path name indicating a storage location.

[0106] The “alteration details” indicates information of alteration performed when a corresponding condition is satisfied. FIG. 7 illustrates an example of stored conceptual information such as “MINF1” in the “alteration details”. However, information actually stored includes specific information indicating a target firmness due to the alteration, such as a pressure value applied to a flexible member corresponding to the firmness after the alteration, or a file path name indicating its storage location.

[0107] In the example of FIG. 7, the condition (condition CD1) identified by the condition ID “CD1” is a condition used for change amount comparison represented by a condition as illustrated in condition details CINF1. In FIG. 7, the condition CD1 indicates that the threshold used in the change amount comparison is “0.5”. The condition CD1 indicates that the firmness is to be altered to the firmness corresponding to the alteration details MINF1 when the condition is satisfied. In FIG. 7, the condition CD1 indicates that when there is a module in which the change amount in the pressure is the threshold “0.5” or more, the firmness of the module is to be altered to the firmness corresponding to the alteration details MINF1.

[0108] Note that the alteration condition information storage unit 124 may store various types of information according to a purpose, not limited to the above.

[0109] Although not illustrated in detail, the user information storage unit 125 according to the embodiment stores user information, being information related to the user. The user information storage unit 125 stores user information including various types of information used to identify the user and various types of information related to attributes of the user. In addition, the user information storage unit 125 stores personal identification information used for personal identification (specification) of the user. For example, the user information storage unit 125 stores a face image of the user used to identify the user. For example, the user information storage unit 125 stores attribute information such as age, sex, residence, place of employment, and interest of the user.

[0110] The user information storage unit 125 may store various types of information according to a purpose, not limited to the above.

[0111] Returning to FIG. 3, and description will continue. The control unit **130** is actualized by execution of programs stored in the information processing apparatus **100** (for example, information processing program such as a determination program according to the present disclosure) by a central processing unit (CPU), a micro processing unit (MPU), or the like, using RAM or the like, as a working area. In addition, the control unit **130** is a controller and is implemented by, for example, an integrated circuit such as an application specific integrated circuit (ASIC) or a field programmable gate array (FPGA).

[0112] As illustrated in FIG. 3, the control unit **130** includes an acquisition unit **131**, an estimation unit **132**, a determination unit **133**, a processing unit **134**, and a transmission unit **135**, and implements or executes a function and an action of information processing described below. The internal configuration of the control unit **130** is not limited to the configuration illustrated in FIG. 3, and may be another configuration as long as it is a configuration that performs information processing described below. Furthermore, the connection relationship of the processing units included in the control unit **130** is not limited to the connection relationship illustrated in FIG. 3, and may be a different connection relationship.

[0113] The acquisition unit **131** acquires various types of information. The acquisition unit **131** acquires various types of information from an external information processing apparatus. The acquisition unit **131** acquires various types of information from the firmness alterable module **10**. The acquisition unit **131** acquires various types of information from another information processing apparatus such as the sensor apparatus **50**.

[0114] The acquisition unit **131** acquires various types of information from the storage unit **120**. The acquisition unit **131** acquires various types of information from the module information storage unit **121**, the history information storage unit **122**, the sensor information storage unit **123**, the alteration condition information storage unit **124**, and the user information storage unit **125**.

[0115] The acquisition unit **131** acquires various types of information estimated by the estimation unit **132**. The acquisition unit **131** acquires various types of information determined by the determination unit **133**. The acquisition unit **131** acquires various types of information processed by the processing unit **134**.

[0116] The acquisition unit **131** acquires sensor information detected by the sensor. The acquisition unit **131** acquires sensor information including pressure information indicating the pressure applied to the surface of the target detected by the pressure sensor. The acquisition unit **131** acquires sensor information including pressure information indicating pressure applied to each of the plurality of regions on the surface of the target. The acquisition unit **131** acquires sensor information including image information captured by the image sensor. The acquisition unit **131** acquires sensor information including image information obtained by imaging the user by the image sensor.

[0117] For example, the acquisition unit **131** acquires, from the firmness alterable module **10**, information indicating the pressure applied to the firmness alterable module **10** detected by a pressure sensor provided in the firmness alterable module **10**. For example, the acquisition unit **131** acquires, from the sensor apparatus **50**, image information captured by the sensor apparatus **50** being an image sensor.

[0118] The estimation unit **132** estimates various types of information. The estimation unit **132** estimates various types of information based on information from an external information processing apparatus or information stored in the storage unit **120**. The estimation unit **132** estimates various types of information from the storage unit **120**. The estimation unit **132** estimates various types of information from the module information storage unit **121**, the history information storage unit **122**, the sensor information storage unit **123**, the alteration condition information storage unit **124**, and the user information storage unit **125**. The estimation unit **132** identifies various types of information. The estimation unit **132** estimates various types of information.

[0119] The estimation unit **132** extracts various types of information. The estimation unit **132** selects various types of information. The estimation unit **132** extracts various types of information

based on information from an external information processing apparatus and information stored in the storage unit **120**. The estimation unit **132** extracts various types of information from the storage unit **120**. The estimation unit **132** extracts various types of information from the module information storage unit **121**, the history information storage unit **122**, the sensor information storage unit **123**, the alteration condition information storage unit **124**, and the user information storage unit **125**.

[0120] The estimation unit **132** extracts various types of information based on the various types of information acquired by the acquisition unit **131**. Furthermore, the estimation unit **132** extracts various types of information based on the various types of information determined by the determination unit **133**. The estimation unit **132** extracts various types of information based on the information that has undergone execution by the processing unit **134**.

[0121] Based on the sensor information acquired by the acquisition unit **131**, the estimation unit **132** estimates a contact mode of an object with respect to the target in which the firmness of the surface is alterable. The estimation unit **132** estimates a contact mode of the object with the surface of the target based on the pressure information. The estimation unit **132** calculates a change amount in the pressure applied by the object to the surface of the target. Based on the pressure information, the estimation unit **132** calculates a change amount of the pressure applied by the object to each of the plurality of regions on the surface of the target.

[0122] Based on the image information, the estimation unit **132** estimates a contact mode of the object with the surface of the target. The estimation unit **132** calculates an intersection between a feature point of the object and the surface of the target by using a moving direction of the object based on the image information. The estimation unit **132** estimates the context related to the object based on the image information. The estimation unit **132** estimates information related to the user based on the image information. The estimation unit **132** estimates a contact mode of the object with respect to a plurality of regions on the surface of the target based on the image information.

[0123] The determination unit **133** determines various types of information. The determination unit **133** identifies various types of information. The determination unit **133** judges various types of information. For example, the determination unit **133** determines various types of information based on information from an external information processing apparatus or information stored in the storage unit **120**. The determination unit **133** determines various types of information based on information from the firmness alterable module **10** and another information processing apparatus such as the sensor apparatus **50**. The determination unit **133** determines various types of information based on information stored in the module information storage unit **121**, the history information storage unit **122**, the sensor information storage unit **123**, the alteration condition information storage unit **124**, or the user information storage unit **125**.

[0124] The determination unit **133** determines various types of information based on the various types of information acquired by the acquisition unit **131**. The determination unit **133** determines various types of information based on the various types of information estimated by the estimation unit **132**. The determination unit **133** determines various types of information based on various types of information executed by the processing unit **134**. The determination unit **133** alters various types of information based on the determination. Various types of information is updated based on the information acquired by the acquisition unit **131**.

[0125] The determination unit **133** determines whether to alter the firmness of the surface of the target according to the contact mode of the object estimated by the estimation unit **132**. The determination unit **133** determines whether to alter the firmness of the surface of the target based on the change amount in the pressure. In a case where the change amount in the pressure is a threshold or more, the determination unit **133** determines to alter the firmness of the surface of the target. In a case where a gesture corresponding to a pressure change pattern satisfies a condition, the determination unit **133** determines to alter the firmness of the surface of the target. The determination unit **133** determines the alteration amount of the firmness of the surface of the target

according to the change amount in the pressure.

[0126] The determination unit **133** determines a region in which the change amount in the pressure satisfies the condition among the plurality of regions on the surface of the target, as the region to undergo firmness alteration. In a case where the distance between the feature point of the object and the intersection is the threshold or more, the determination unit **133** determines to alter the firmness of the surface of the target. The determination unit **133** determines the alteration amount of the firmness of the surface of the target according to the moving speed of the object.

[0127] In a case where the context estimated by the estimation unit **132** satisfies the condition, the determination unit **133** determines to alter the firmness of the surface of the target. The determination unit **133** determines the alteration amount in the firmness of the surface of the target according to the context. The determination unit **133** determines whether to alter the firmness of the surface of the target based on the information related to the user estimated by the estimation unit **132**. The determination unit **133** determines a region in which the contact mode satisfies the condition among the plurality of regions on the surface of the target, as a region to undergo firmness alteration. The determination unit **133** determines a force used to press a flexible member **31** by a pressing mechanism **32**.

[0128] The processing unit **134** executes various types of processing. The processing unit **134** executes various types of information based on information from an external information processing apparatus or information stored in the storage unit **120**. The processing unit **134** executes various types of information based on information from the firmness alterable module **10** and another information processing apparatus such as the sensor apparatus **50**. The processing unit **134** executes various types of information based on information stored in the module information storage unit **121**, the history information storage unit **122**, the sensor information storage unit **123**, the alteration condition information storage unit **124**, or the user information storage unit **125**.

[0129] The processing unit **134** executes various types of information processing based on the various types of information acquired by the acquisition unit **131**. The processing unit **134** executes various types of information based on the various types of information estimated by the estimation unit **132**. The processing unit **134** executes various types of information based on the various types of information determined by the determination unit **133**.

[0130] The processing unit **134** executes processing related to the alteration of the firmness of the firmness alterable module **10**. The processing unit **134** executes processing of altering the firmness of the firmness alterable module **10** to the firmness determined by the determination unit **133**. The processing unit **134** controls the firmness of the firmness alterable module **10** by instructing the transmission unit **135**. The processing unit **134** instructs the transmission unit **135** to transmit information indicating firmness to the firmness alterable module **10**. The processing unit **134** controls the force of pressing the flexible member **31** by the pressing mechanism **32** to execute processing of altering the firmness of the firmness alterable module **10**.

[0131] The transmission unit **135** provides various types of information to an external information processing apparatus. The transmission unit **135** transmits various types of information to an external information processing apparatus. For example, the transmission unit **135** transmits various types of information to the firmness alterable module **10** and another information processing apparatus such as the sensor apparatus **50**. The transmission unit **135** provides the information stored in the storage unit **120**. The transmission unit **135** transmits the information stored in the storage unit **120**.

[0132] The transmission unit **135** provides various types of information based on information from the firmness alterable module **10** and another information processing apparatus such as the sensor apparatus **50**. The transmission unit **135** provides various types of information based on the information stored in the storage unit **120**. The transmission unit **135** provides various types of information based on information stored in the module information storage unit **121**, the history information storage unit **122**, the sensor information storage unit **123**, the alteration condition

information storage unit **124**, or the user information storage unit **125**.

[0133] The transmission unit **135** transmits various types of information based on the various types of information acquired by the acquisition unit **131**. The transmission unit **135** transmits various types of information based on the various types of information estimated by the estimation unit **132**. The transmission unit **135** transmits various types of information based on the various types of information determined by the determination unit **133**. The transmission unit **135** notifies various types of information based on various types of information executed by the processing unit **134**. The transmission unit **135** transmits various types of information to the firmness alterable module **10** in response to an instruction from processing unit **134**.

[0134] The transmission unit **135** transmits instruction information instructing the firmness alterable module **10** to alter the firmness to the firmness alterable module **10** in accordance with an instruction from the processing unit **134**. The transmission unit **135** transmits instruction information instructing the firmness alterable module **10** to alter the firmness to the firmness alterable module **10** in accordance with an instruction from the processing unit **134**. The transmission unit **135** transmits control information of controlling the firmness alterable module **10** to the firmness alterable module **10** in accordance with an instruction from the processing unit **134**. The transmission unit **135** transmits, to the firmness alterable module **10**, information designating a force used to press the flexible member **31** by the pressing mechanism **32**.

1-4. Configuration of Firmness Alterable Module According to Embodiment

[0135] The following will describe a configuration of the firmness alterable module **10**, which is an example of an output apparatus that executes output processing according to the embodiment. FIG. **8** is a diagram illustrating a configuration example of a firmness alterable module according to the embodiment of the present disclosure.

[0136] When pressure is detected, the firmness alterable module **10** may include a pressure sensor. For example, the firmness alterable module **10** includes a pressure sensor that detects a pressure applied to a surface **11** (an upper surface in FIG. **8**). As long as the pressure applied to the surface **11** can be detected, a sensor (pressure sensor) that detects the pressure applied to the surface **11** may be provided outside the firmness alterable module **10**.

[0137] As illustrated in FIG. **8**, the firmness alterable module **10** includes a surface forming member **20**, a softness/firmness controller **30**, and a cover portion **40**.

[0138] The surface forming member **20** has a structure for maintaining a surface shape. FIG. **8** illustrates a case where the surface forming member **20** has a structure that maintains a planar shape. However, the surface forming member **20** may have any structure as long as the surface shape is maintained. For example, the surface forming member **20** may have a structure that maintains a curved surface shape.

[0139] The surface forming member **20** includes a coupling portion **21** and a block portion **22**. The coupling portion **21** is a sheet-like member that is flexible in a direction intersecting the surface **11**. The coupling portion **21** is a sheet-like member disposed along the surface **11** and flexible in a direction intersecting the surface **11**. The coupling portion **21** is formed of a material that permits bending but does not easily permit elongation. The coupling portion **21** is formed of a material having high flexibility and low elasticity. For example, the coupling portion **21** uses a plastic film, a nonwoven fabric, a nylon fabric, or the like.

[0140] The block portion **22** includes a plurality of blocks **220**. The block portion **22** includes a plurality of blocks **220** aligned along a back surface which is a surface on a side opposite to the surface of the coupling portion **21** and coupled to the back surface.

[0141] The block **220** is desirably a rigid body that does not deform. For example, the block **220** may use rigid plastic, metal, wood, or the like. The block **220** has a first surface **221** and a second surface **222**. Each of the plurality of blocks **220** is a hexahedron having the second surface **222**, being an opposing surface facing the first surface **221**, coupled to the coupling portion **21**. Surfaces other than the first surface **221** and the second surface **222** are disposed adjacent to surfaces of the

other blocks **220**.

[0142] As illustrated in FIG. **9**, the plurality of blocks **220** are disposed side by side over the back surface of the coupling portion **21**. FIG. **9** is a plan view illustrating a main part of the firmness alterable module. FIG. **9** is a plan view of the surface forming member **20** from the back surface side of the coupling portion **21**. In order to indicate that there is a clearance CR between the blocks **220**, FIG. **9** explicitly illustrates a gap between the blocks **220**. However, the clearance CR is desirably close to 0 mm. For example, the closer the clearance CR between the blocks **220** to 0 mm, the more the undesired deformation of the surface forming member **20** can be suppressed as compared with a case where the clearance CR is large.

[0143] Here, an example of deformation of the surface forming member **20** when a force is applied to the surface forming member **20** will be described with reference to FIGS. **10** and **11**. FIGS. **10** and **11** are side views illustrating main parts of the firmness alterable module.

[0144] Specifically, FIG. **10** illustrates a case where a force is applied from the block portion **22** side of the surface forming member **20**, that is, from the inner side (softness/firmness controller **30** side) of the firmness alterable module **10**. When a force is applied to the surface forming member **20** from the inner side of the firmness alterable module **10** in this manner, the blocks **220** interfere with each other to suppress deformation of the surface forming member **20**. That is, when a force is applied to the surface forming member **20** from the inner side of the firmness alterable module **10**, the shape of the surface **11** of the firmness alterable module **10** is maintained.

[0145] For example, in a case where the module does not include the surface forming member **20** which is a component for maintaining the surface shape of the surface, the upper surface of the module would protrude at the top at the time of compression. In contrast, by disposing the surface forming member **20** between the cover portion **40** and the flexible member **31**, the firmness alterable module **10** is capable of maintaining the surface shape at the time of compression.

[0146] FIG. **11** illustrates a case where a force is applied from the coupling portion **21** side of the surface forming member **20**, that is, from the outer side (surface **11** side) of the firmness alterable module **10**. When a force is applied to the surface forming member **20** from the outer side of the firmness alterable module **10** in this manner, the surface forming member **20** easily deforms with no mutual interference of the blocks **220**. That is, when a force is applied to the surface forming member **20** from the outer side of the firmness alterable module **10**, the shape of the surface **11** of the firmness alterable module **10** is deformed.

[0147] The method of forming the surface forming member **20** can adopt any method. For example, the surface forming member **20** may be formed by adhesive forming. For example, the surface forming member **20** may be formed by forming the block portion **22** and then bonding the block portion **22** to the material of the coupling portion **21**.

[0148] For example, the surface forming member **20** may be formed by integral forming. For example, the surface forming member **20** may be integrally formed with tough nylon. In this case, the coupling portion **21** and the block portion **22** may be formed using nylon. For example, the surface forming member **20** may be integrally formed by thickening the block portion **22** and thinning the coupling portion **21** as much as possible. The surface forming member **20** may be integrally formed by a multi-material 3D printer. In this case, it is allowable to use a rigid material for the block portion **22** and use a soft material for the coupling portion **21** and combine them to be integrally formed with each other. The above is merely an example, and the surface forming member **20** may be formed by any forming method as long as the block portion **22** having the plurality of blocks **220** is coupled to one surface (back surface) of the coupling portion **21**.

[0149] The softness/firmness controller **30** has a function of altering firmness. The softness/firmness controller **30** is disposed along one surface of each of the plurality of blocks **220** with the block portion **22** interposed between the softness/firmness controller **30** and the coupling portion **21**. The softness/firmness controller **30** includes the flexible member **31** and the pressing mechanism **32**.

[0150] The flexible member **31** is a flexible material to be compressed. The flexible member **31** is formed with a flexible material. The flexible member **31** is desirably formed by using a material having elasticity and high shrinkability. For example, the flexible member **31** may be a flexible urethane foam, a spring, or the like.

[0151] The flexible member **31** is disposed along one surface of each of the plurality of blocks **220**. For example, the flexible member **31** is disposed along the first surface **221** of each of the plurality of blocks **220** with the block portion **22** interposed between the flexible member **31** and the coupling portion **21**. The flexible member **31** is compressed by being pressed toward the surface forming member **20** by the pressing mechanism **32**.

[0152] The pressing mechanism **32** has a function of pressing the flexible member **31**. The pressing mechanism **32** presses the flexible member **31** toward the block portion **22** from a position sandwiching the block portion **22** and the flexible member **31**. The pressing mechanism **32** includes a movable portion **321**, a drive unit **322**, a shaft unit **323**, and a support unit **324**.

[0153] The movable portion **321** has a plate shape along the flexible member **31** at a position sandwiching the block portion **22** and the flexible member **31**. The drive unit **322** moves the movable portion **321** in a direction intersecting the surface. The shaft unit **323** is attached to the movable portion **321** and transmits a force generated by the drive unit **322** to the movable portion **321**. The support unit **324** functions as a support of the drive unit **322**. For example, the support unit **324** supports the drive unit **322** and holds the position of the drive unit **322**.

[0154] For example, the drive unit **322** includes a drive mechanism such as an actuator or a motor, and moves the shaft unit **323** forward and backward with respect to the flexible member **31** so as to move the movable portion **321** forward and backward with respect to the flexible member **31**. With this configuration, the drive unit **322** alters the pressure on the flexible member **31** applied by the movable portion **321**. The drive unit **322** is driven under the control of the information processing apparatus **100**, and alters the pressing mode of the flexible member **31** to alter the firmness of the surface **11** of the firmness alterable module **10**.

[0155] For example, when a linear motion motor having a screw lead of 1 millimeter (mm) and the number of steps of **200** is used for the drive unit **322**, the firmness alterable module **10** operates as follows. Hereinafter, the frequency is assumed to be 1000 to 5000 pulses per second (pps).

[0156] For example, in the case of driving with 1000 pps, the drive unit **322** can be driven at 5 millimeter/second (mm/s), and can compress the flexible member **31** such as a piece of sponge by 60 mm at 15 second(s). Further, for example, in the case of driving with 5000 pps, the drive unit **322** can be driven at 25 mm/s, and can compress the flexible member **31** of the sponge by 60 mm at 2.4 seconds(s).

[0157] Although the above-described example has described an exemplary case where the softness/firmness controller **30** is constituted with the flexible member **31** and the pressing mechanism **32**, the softness/firmness controller **30** can adopt any configuration as long as the firmness can be altered. For example, the softness/firmness controller **30** may be actualized by a powder. In this case, the softness/firmness controller **30** may be configured to be able to alter the firmness also by using jamming transition of the powder. For example, the softness/firmness controller **30** using the powder is softened by mixing the powder and air. In addition, for example, the softness/firmness controller **30** using the powder comes into a firm state by removing air in a state where the powder is pressed against the upper surface.

[0158] The cover portion **40** is a cover material that covers the firmness alterable module **10**. The cover portion **40** is formed of a material having low elasticity. For example, the cover portion **40** is formed by using a material that is difficult to stretch, such as cloth or leather. The cover portion **40** may be a mesh (material). The cover portion **40** is disposed so as to cover the coupling portion **21**, the block portion **22**, and the softness/firmness controller **30**. Another object such as a person comes into contact with the opposite surface (corresponding to the surface **11** in FIG. **8**) of the surface of the cover portion **40** running along the coupling portion **21**.

[1-4-1. Configuration Example of Object Using Firmness Alterable Module]

[0159] The firmness alterable module **10** used for the object **2** provides an interface that alters the firmness of the surface of the object **2** with the above-described configuration. Note that any number of the firmness alterable modules **10** may be used for the object **2**. For example, the object **2** may have a configuration in which one firmness alterable module **10** is used to alter the firmness of the entire surface of the object **2**.

[0160] Furthermore, for example, the object **2** may be a mechanism in which the firmness of the surface of the object **2** is partially changed by arranging the firmness alterable modules **10** in an array. For example, as illustrated in FIG. **1**, the object **2** may have a configuration in which a plurality of firmness alterable modules **10** are disposed side by side in a planar shape so as to partially alter the firmness of the surface of the object **2**. FIG. **12** illustrates an example of arrangement of the plurality of firmness alterable modules **10**.

[0161] FIG. **12** is a diagram illustrating an arrangement example of a plurality of firmness alterable modules. The diagram illustrates a state in which five firmness alterable modules **10-11** to **10-15** are disposed side by side. Although FIG. **12** illustrates only the five modules, the firmness alterable modules **10** more than five may be disposed side by side, or four or less of the firmness alterable modules **10** may be disposed side by side. For example, by arranging the firmness alterable modules **10**, specifically each of the firmness alterable modules **10-11** to **10-15** illustrated in FIG. **12**, side by side also in the depth direction of the page, a plurality of the firmness alterable modules **10** is to be disposed side by side in a planar shape as illustrated in FIG. **1**.

1-5. Processing Example

[0162] Hereinafter, specific examples of various types of processing will be described with reference to FIGS. **13** to **26**. In the following description, description of points similar to the above-described contents will be appropriately omitted.

[1-5-1. Information Processing Procedure According to Embodiment]

[0163] Next, an information processing procedure according to the embodiment will be described with reference to FIG. **13**. FIG. **13** is a flowchart illustrating a procedure of information processing according to the embodiment of the present disclosure. Specifically, FIG. **13** is a flowchart illustrating a procedure of information processing performed by the information processing apparatus **100**.

[0164] As illustrated in FIG. **13**, the information processing apparatus **100** acquires sensor information detected by a sensor (step **S101**). For example, the information processing apparatus **100** acquires sensor information indicating a change in pressure corresponding to the firmness alterable module **10** detected by a pressure sensor provided in the firmness alterable module **10**.

[0165] Based on the sensor information, the information processing apparatus **100** estimates a contact mode of an object with respect to a target in which the firmness of the surface is alterable (step **S102**). For example, the information processing apparatus **100** calculates the change amount in the pressure to the firmness alterable module **10** based on the change in pressure corresponding to the firmness alterable module **10**.

[0166] Subsequently, the information processing apparatus **100** determines whether to alter the firmness of the surface of the target according to the contact mode of the object (step **S103**). In a case where the change amount in the pressure to the firmness alterable module **10** is a threshold or more, the information processing apparatus **100** determines to alter the firmness of the firmness alterable module **10**.

[1-5-2. Processing Example According to Pressure]

[0167] The outline of the information processing procedure by the information processing apparatus **100** has been described with FIG. **13**. Hereinafter, specific examples will be described with reference to FIGS. **14** to **26**. For example, the information processing apparatus **100** may alter the firmness of the module according to the change in pressure. For example, in a case where a certain weight (pressure) or more is detected for the object **2**, the information processing apparatus

100 may alter the firmness of the firmness alterable module **10** of the object **2**. For example, in a case where another object such as a coffee cup or a book is placed on the object **2**, the information processing apparatus **100** may alter the firmness of the firmness alterable module **10** such that a position on the object **2** where the another object is placed becomes a firm surface like a side table. In contrast, regarding a position where a person sits, the information processing apparatus **100** may alter the firmness of the firmness alterable module **10** so that the position becomes a soft surface like a sofa.

[0168] The information processing system **1** performs sensing by pressure, and alters firmness of a surface of an object according to a sensing result. The information processing system **1** determines a deformation position. For example, the information processing system **1** determines a module to be deformed based on pressure distribution. The information processing system **1** determines firmness. For example, the information processing system **1** calculates the deformation amount from the pressure based on a preset deformation amount. The information processing system **1** alters firmness. For example, the information processing system **1** drives an actuator. In this manner, in the case of pressure sensing, the information processing system **1** determines the position to be deformed and the firmness of each module based on a timing at which the module and the object come into contact with each other, and drives each module.

[0169] Hereinafter, a specific example of interaction by pressure sensing will be described with reference to FIG. **14**. FIG. **14** is a flowchart illustrating a procedure of processing according to pressure.

[0170] As illustrated in FIG. **14**, in a case where the switch is not turned ON (step S201: No), the information processing apparatus **100** drives the actuator until all the modules are in firm states (step S202). For example, in a case where the switch for firmness alteration is OFF and the mode is not the firmness alteration mode, the information processing apparatus **100** controls the softness/firmness controller **30** until all the firmness alterable modules **10** of the object **2**, which is the target, are in firm states. The information processing apparatus **100** maintains the state set in step S202 and stops firmness alteration.

[0171] The firm state herein may be the firmest state in a range alterable by the softness/firmness controller **30**. Furthermore, although FIG. **14** is an exemplary case where the firm state is the initial setting state (default state), the default state is not limited to the firm state, and may be a soft (soft) state or an intermediate firmness state between the firm state and the soft state. The soft state here may be the softest state in the range alterable by the softness/firmness controller **30**.

[0172] In a case where the switch is turned ON (step S201: Yes), the information processing apparatus **100** measures the pressure of each module (step S203). For example, in a case where the switch for firmness alteration is ON and the mode is the firmness alteration mode, the information processing apparatus **100** acquires information indicating the pressure of each of the firmness alterable modules **10** from the pressure sensor that detects the pressure of each of the firmness alterable modules **10**.

[0173] Subsequently, the information processing apparatus **100** calculates a change amount in the pressure of each module (step S204). For example, the information processing apparatus **100** calculates the change amount of the pressure of each firmness alterable module **10** to estimate the contact mode with the object **2** equipped with the firmness alterable module **10**.

[0174] Subsequently, the information processing apparatus **100** judges whether there is a module in which the pressure is a threshold or more (step S205). For example, the information processing apparatus **100** judges whether there is a firmness alterable module **10** in which the pressure is a threshold or more.

[0175] In a case where there is no module whose pressure is the threshold or more (step S205: No), the information processing apparatus **100** drives the actuator until all the modules are in firm states (step S206). For example, in a case where there is no firmness alterable module **10** in which the pressure is the threshold or more, the information processing apparatus **100** controls the

softness/firmness controller **30** until all the firmness alterable modules **10** of the object **2** are in firm states. Subsequently, the information processing apparatus **100** returns to step **S201** and repeats the processing.

[0176] In a case where there is a module in which the pressure is the threshold or more (step **S205**: Yes), the information processing apparatus **100** judges whether there is a module in which the change amount is the threshold or more (step **S207**). For example, the information processing apparatus **100** judges whether there is a firmness alterable module **10** in which the pressure change amount is a threshold or more.

[0177] In a case where there is no module in which the change amount is the threshold or more (step **S207**: No), the information processing apparatus **100** returns to step **S201** and repeats the processing. For example, in a case where there is no firmness alterable module **10** in which the change amount in the pressure is the threshold or more, the information processing apparatus **100** returns to step **S201** and repeats the processing without altering the firmness of the object **2**.

[0178] In a case where there is a module in which the change amount is the threshold or more (step **S207**: Yes), the information processing apparatus **100** specifies a module to be deformed (step **S208**). For example, the information processing apparatus **100** determines the firmness alterable module **10** in which the pressure change amount is a threshold or more as a target module to undergo firmness alteration.

[0179] Subsequently, the information processing apparatus **100** determines softness (step **S209**). For example, the information processing apparatus **100** determines the softness of the firmness alterable module **10** determined as the target module. The information processing apparatus **100** determines to decrease the firmness of the firmness alterable module **10** such that the greater the pressure change amount, the more the firmness of the firmness alterable module **10** is to be decreased.

[0180] Subsequently, the information processing apparatus **100** drives the actuator (step **S210**). For example, the information processing apparatus **100** controls the softness/firmness controller **30** according to the determined firmness of the firmness alterable module **10**. Subsequently, the information processing apparatus **100** returns to step **S201** and repeats the processing.

[0181] As described above, in an interaction case using pressure sensing, the information processing apparatus **100** executes various types of processing according to ON and OFF states of the switch. For example, in a case where the switch is ON, the information processing apparatus **100** executes the following processing.

[0182] The information processing apparatus **100** measures the pressure of each module and calculates the change amount in the pressure. In a case where there is a module in which the pressure is the threshold or more and the change amount also exceeds the threshold, the information processing apparatus **100** sets the module as a target module to undergo deformation. The information processing apparatus **100** determines softness. In this case, the information processing apparatus **100** may set predetermined softness or may calculate softness with respect to pressure.

[0183] The information processing apparatus **100** drives the actuator to soften the module. Furthermore, in a case where a module that is already soft is excluded from the target, the information processing apparatus **100** may return the module to the firm state. In a case where there is no module in which the pressure is the threshold or more, the information processing apparatus **100** drives the actuator so set all the modules to firm states.

[0184] Furthermore, in a case where the switch is OFF, the information processing apparatus **100** drives the actuator to set all the modules to firm states. Furthermore, the information processing apparatus **100** may specify a target object using a camera or a tag and set a threshold according to the target object. In this case, the information processing apparatus **100** may lower the threshold of the pressure or the pressure change amount for a child, for example.

[0185] Note that the processing using the pressure described above is merely an example, and the

information processing apparatus **100** may perform various types of processing using pressure information. For example, the information processing apparatus **100** may alter the firmness of the firmness alterable module **10** according to the pressure of another adjacent firmness alterable module **10**. This point will be described with reference to FIG. **15**. FIG. **15** is a diagram illustrating an example of alteration according to an adjacent module. FIG. **15** illustrates a pressure distribution diagram when the object **2** equipped with the firmness alterable module **10** is viewed from above. For example, the information processing apparatus **100** softens the firmness alterable module **10** in which the pressure is the threshold or more.

[0186] In FIG. **15**, the darker the color of hatching, the larger the pressure value. For example, FIG. **15** illustrates a case where the pressures of eight modules, namely, modules M**13**, M**15**, M**23** to M**25**, and M**33** to M**35** of the object **2** are equal to or more than the threshold as the condition for being the target module. Here, all the adjacent modules of the module M**14**, namely, the three modules M**13**, M**15**, and M**24**, have pressures equal to or more than the threshold of the condition for being the target module. Therefore, the information processing apparatus **100** also determines the module M**14**, being a module having three or more adjacent side modules with alteration, a target module to undergo firmness alteration. In this manner, the information processing apparatus **100** may set the module M**14** to be a module to be deformed because the pressures of the three surrounding side modules are the threshold or above even though the pressure of the module M**14** itself is the threshold or less.

[0187] Furthermore, the information processing apparatus **100** may alter the firmness of the module according to pressure change patterns. As an interaction case by pressure sensing, the information processing apparatus **100** may perform gesture recognition by a pressure change pattern and alter the firmness of the module according to the gesture. In this case, the information processing apparatus **100** may alter the firmness of the module when the gesture is detected.

[0188] For example, in a case where a gesture of pushing a surface of the object **2** is detected, the information processing apparatus **100** may soften, that is, decrease the firmness of a module corresponding to the surface. Furthermore, in a case where a gesture of pushing a soft surface of the object **2** is detected, the information processing apparatus **100** may increase the firmness of the module corresponding to the surface. Furthermore, the information processing apparatus **100** may learn a pressure pattern at the time of rising from the object **2** such as a cushion and increase the firmness of the module corresponding to the surface of the object **2** at the time of rising.

[0189] This point will be described with reference to FIG. **16**. FIG. **16** is a flowchart illustrating a procedure of processing according to a pressure change pattern. Hereinafter, description of the points similar to the points described with FIG. **14** will be omitted as appropriate.

[0190] As illustrated in FIG. **16**, in a case where the switch is not turned ON (step S**301**: No), the information processing apparatus **100** drives the actuator until all the modules are in firm states (step S**302**). The information processing apparatus **100** maintains the state set in step S**302** and stops firmness alteration.

[0191] In a case where the switch is turned ON (step S**301**: Yes), the information processing apparatus **100** measures the pressure of each module (step S**303**). Subsequently, the information processing apparatus **100** calculates a change amount of the pressure of each module (step S**304**).

[0192] Subsequently, the information processing apparatus **100** estimates a gesture from the change pattern (step S**305**). For example, the information processing apparatus **100** estimates a gesture based on a pressure change pattern of each firmness alterable module **10** to estimate a contact mode with the object **2** equipped with the firmness alterable module **10**.

[0193] Subsequently, the information processing apparatus **100** judges whether the gesture has been recognized (step S**306**). For example, the information processing apparatus **100** judges whether the pressure change pattern of each firmness alterable module **10** matches the change pattern corresponding to the gesture, thereby judging whether the gesture has been recognized. For example, in a case where the pattern of the change in pressure of each firmness alterable module **10**

matches the change pattern corresponding to the gesture, the information processing apparatus **100** judges that the gesture has been recognized. For example, in a case where the pattern of the change in pressure of each firmness alterable module **10** does not match the change pattern corresponding to the gesture, the information processing apparatus **100** judges that the gesture has not been recognized.

[0194] In a case where the gesture has not been recognized (step **S306**: No), the information processing apparatus **100** returns to step **S301** and repeats the processing.

[0195] In a case where the gesture has been recognized (step **S306**: Yes), the information processing apparatus **100** specifies a module to be deformed (step **S307**). For example, the information processing apparatus **100** determines the firmness alterable module **10** in which the pressure change amount is a threshold or more as a target module to undergo firmness alteration.

[0196] Subsequently, the information processing apparatus **100** determines softness (step **S308**). For example, the information processing apparatus **100** determines the softness of the firmness alterable module **10** determined as the target module. The information processing apparatus **100** determines to decrease the firmness of the firmness alterable module **10** such that the greater the pressure change amount, the more the firmness of the firmness alterable module **10** is to be decreased.

[0197] Subsequently, the information processing apparatus **100** drives the actuator (step **S309**). For example, the information processing apparatus **100** controls the softness/firmness controller **30** according to the determined firmness of the firmness alterable module **10**. Subsequently, the information processing apparatus **100** returns to step **S301** and repeats the processing.

1-5-3. Processing Example According to Object Detection

[0198] Furthermore, the information processing apparatus **100** may alter the firmness of the module using sensor information (image) detected by the camera.

[0199] The information processing system **1** performs sensing by a camera, and alters firmness of a surface of an object according to a sensing result. The information processing system **1** estimates a target object. For example, the information processing system **1** estimates a target object using object recognition. The information processing system **1** determines a deformation position. For example, the information processing system **1** determines a module to be deformed from the feature point and its displacement. The information processing system **1** determines firmness. For example, the information processing system **1** determines firmness based on a library of deformation amounts for a target object. For example, the information processing system **1** may calculate the deformation amount from the speed at the time of contact. Furthermore, for example, the information processing system **1** may calculate the deformation amount from the pressure. The information processing system **1** alters firmness. For example, the information processing system **1** drives an actuator. In this manner, in the case of sensing by the camera, the information processing system **1** can start interaction before the module and the object come into contact with each other. For example, the information processing system **1** can alter the firmness even before the module and the object come into contact with each other, can later the firmness according to the object or the context, and can estimate the speed at the time of contact.

[0200] Hereinafter, an outline of an interaction case by camera sensing will be described with reference to FIG. **17**. FIG. **17** is a conceptual diagram illustrating an example of alteration according to object detection. FIG. **17** illustrates a state in which a user **U** is about to sit on the object **2** which is a sofa. The information processing apparatus **100** extracts feature points of the user **U** using an image obtained by imaging the user **U**. The information processing apparatus **100** extracts the feature point of the user **U** appropriately using a certain technology, which may be any technology as long as the feature point of the object can be extracted by camera sensing, and detailed description will be omitted. Subsequently, the information processing apparatus **100** calculates a motion vector of the extracted feature point of the user **U** using the image obtained by imaging the user **U**. The information processing apparatus **100** calculates the motion vector of the

feature point of the user U appropriately using a certain technology, which may be any technology as long as the motion vector of the feature point of the object can be calculated by camera sensing, and detailed description will be omitted.

[0201] In FIG. 17, the information processing apparatus 100 calculates a motion vector V1 of the feature point of the user U. Subsequently, the information processing apparatus 100 determines to soften, that is, decrease the firmness of the module in which a distance between an intersection on the module with the motion vectors from the feature point of the user U and the feature point is a predetermined value or less. For example, the information processing apparatus 100 determines an intersection on the surface of the object 2 with a motion vector from a feature point of the user U by using an image obtained by imaging the user U. Subsequently, the information processing apparatus 100 calculates a distance (also referred to as a “distance between intersections”) between each feature point of the user U and an intersection on the surface of the object 2 with the motion vector from each feature point by using the image obtained by imaging the user U. The information processing apparatus 100 determines to soften, that is, decrease the firmness of the module of the object 2, including the intersection where the distance between the intersections is a predetermined value or less. In FIG. 17, the information processing apparatus 100 determines to soften, that is, decrease the firmness of the hatched four modules from the right.

[0202] In this manner, the information processing apparatus 100 alters firmness when a target object such as the user U is detected. For example, in a case where a person tries to sit on the object 2, the information processing apparatus 100 alters the firmness of the module corresponding to the surface of the object 2 so that the surface becomes a soft surface like a sofa. Furthermore, in the case of an unstable object having a curved surface on the contact surface, the information processing apparatus 100 alters the firmness of the module corresponding to the surface of the object 2 so as to be a soft surface.

[0203] Hereinafter, an example of a flow of processing according to object detection will be described with reference to FIG. 18. FIG. 18 is a flowchart illustrating a procedure of processing according to object detection. Hereinafter, description of the points similar to the points described with FIG. 14, etc. will be omitted as appropriate.

[0204] As illustrated in FIG. 18, in a case where the switch is not turned ON (step S401: No), the information processing apparatus 100 drives the actuator until all the modules are in firm states (step S402). The information processing apparatus 100 maintains the state set in step S402 and stops firmness alteration.

[0205] In a case where the switch is ON (step S401: Yes), the information processing apparatus 100 perform object detection (step S403). The information processing apparatus 100 appropriately performs object detection using a certain technology, which may be any technology as long as object detection can be performed using an image, and detailed description thereof will be omitted. For example, in a case where the switch for firmness alteration is ON and the mode is the firmness alteration mode, the information processing apparatus 100 performs object detection using an image captured by the sensor apparatus 50 being a camera.

[0206] Subsequently, the information processing apparatus 100 judges whether a target object has been detected (step S404). In a case where the target object has not been detected (step S404: No), the information processing apparatus 100 returns to step S401 and repeats the processing.

[0207] When a target object is detected (step S404: Yes), the information processing apparatus 100 calculates a motion vector of each feature point (step S405). For example, in a case where a target object is detected, the information processing apparatus 100 extracts a feature point of the target object and calculates a motion vector of each extracted feature point.

[0208] Subsequently, the information processing apparatus 100 calculates an intersection between each motion vector and the module upper surface (step S406). For example, the information processing apparatus 100 calculates the intersection of the motion vector from the feature point of the target object with the upper surface of the module of the object 2.

[0209] The information processing apparatus **100** judges whether there is a feature point whose distance to the module upper surface is within a threshold (step **S407**). For example, the information processing apparatus **100** judges whether there is a feature point whose distance from the feature point of the target object to the intersection on the upper surface of the module of the object **2** corresponding to the feature point is within the threshold.

[0210] In a case where there is no feature point whose distance to the module upper surface is within the threshold (step **S407**: No), the information processing apparatus **100** drives the actuator until all the modules are in firm states (step **S408**). For example, in a case where there is no feature point whose distance from the feature point of the target object to the intersection on the upper surface of the module of the object **2** corresponding to the feature point is within the threshold, the information processing apparatus **100** controls the softness/firmness controller **30** until all the firmness alterable modules **10** of the object **2** are in firm states. Subsequently, the information processing apparatus **100** returns to step **S401** and repeats the processing.

[0211] In a case where there is a feature point whose distance to the module upper surface is within the threshold (step **S407**: Yes), the information processing apparatus **100** specifies a module to be deformed (step **S409**). For example, the information processing apparatus **100** determines the firmness alterable module **10** including an intersection whose distance from the feature point of the target object is within a threshold as a target module to undergo firmness alteration.

[0212] Subsequently, the information processing apparatus **100** determines softness (step **S410**). For example, the information processing apparatus **100** determines the softness of the firmness alterable module **10** determined as the target module. The information processing apparatus **100** determines to decrease the firmness of the firmness alterable module **10** such that the higher the moving speed of the target object, the more the firmness of the firmness alterable module **10** being a target module is to be decreased.

[0213] Subsequently, the information processing apparatus **100** drives the actuator (step **S411**). For example, the information processing apparatus **100** controls the softness/firmness controller **30** according to the determined firmness of the firmness alterable module **10**. Subsequently, the information processing apparatus **100** returns to step **S401** and repeats the processing.

[0214] As described above, in an interaction case using camera sensing, the information processing apparatus **100** executes various types of processing according to ON and OFF states of the switch. For example, in a case where the switch is ON, the information processing apparatus **100** executes the following processing.

[0215] The information processing apparatus **100** performs object estimation. In a case where an object being a target is detected, the information processing apparatus **100** calculates a motion vector of each feature point of the object. For example, when having detected a target object such as a person or an object having a curved surface for which the module is to be softened, the information processing apparatus **100** calculates a motion vector of each feature point of the object.

[0216] The information processing apparatus **100** calculates an intersection between the motion vector and the module upper surface. In a case where there is a feature point whose distance to the module upper surface is within a threshold, the information processing apparatus **100** defines a module including an intersection of a motion vector of the feature point and the module upper surface, as a target module to undergo deformation. The information processing apparatus **100** determines softness. The information processing apparatus **100** may prepare a library of softness for the target object in advance and refer to the value. In a case where a pressure sensor is included in the module, the information processing apparatus **100** may calculate softness with respect to pressure.

[0217] The information processing apparatus **100** may alter the softness in consideration of the moving speed of the feature point. When the module to be deformed is determined, the information processing apparatus **100** may drive the actuator to set the module in a soft state. Furthermore, in a case where a module that is already soft is excluded from the target, the information processing

apparatus **100** may return the module to the firm state. In a case where there is no feature point whose distance to the module upper surface is within the threshold, the information processing apparatus **100** may drive the actuator so that all the modules are in firm states.

[0218] Furthermore, in a case where the switch is OFF, the information processing apparatus **100** drives the actuator to set all the modules to firm states.

[1-5-4. Processing Example According to Context]

[0219] The information processing apparatus **100** may alter the firmness of the module according to the context. First, an outline of an interaction case according to a context will be described with reference to FIGS. **19** and **20**. FIG. **19** is a conceptual diagram illustrating an example of alteration according to a context. FIG. **20** is a conceptual diagram illustrating an example of alteration according to a context.

[0220] FIG. **19** illustrates a state in which a user U is in a working state and brings their arms into contact with the object **2** being a desk. In addition, FIG. **20** illustrates a state in which the user U is in a state of facing down on the desk, that is, in a state in which the user U sleeps, with their arms being in contact with the object **2** being a desk. Using an image obtained by imaging the user U, the information processing apparatus **100** estimates an intended use of the object **2** by the user U as a context. In a case where the image in which the user U is in the state of FIG. **19** is acquired, the information processing apparatus **100** estimates that the object **2**, being a desk, is a context used for work based on bone information detected by bone detection as illustrated in a bone B1 of the user U. In another case where the image in which the user U is in the state of FIG. **20** is acquired, the information processing apparatus **100** estimates that the object **2**, being a desk, is a context used for rest based on bone information detected by bone detection as illustrated in a bone B2 of the user U.

[0221] For example, after a target context has been detected, the information processing apparatus **100** alters firmness. For example, as illustrated in FIG. **19**, the information processing apparatus **100** alters the firmness of the module such that the entire surface of the object **2** will be a firm surface like a desk when the user U works. Furthermore, for example, in a case where the user U tries to sleep facing down as illustrated in FIG. **20**, the information processing apparatus **100** alters the firmness of the module so that the surface with which the user U comes in contact becomes a soft surface. In FIG. **20**, the information processing apparatus **100** determines to soften, that is, decrease the firmness of the five hatched central modules. Furthermore, in a case where the user U tries to rest elbows, the information processing apparatus **100** may alter the firmness of the module so that a portion where the elbow of the user U touches becomes a soft surface.

[0222] Hereinafter, an example of a flow of processing according to the context will be described with reference to FIG. **21**. FIG. **21** is a flowchart illustrating a procedure of processing according to a context. Note that description of points similar to the description in FIGS. **14**, **18**, and the like will be omitted as appropriate.

[0223] As illustrated in FIG. **21**, the information processing apparatus **100** performs bone detection (step S501). The information processing apparatus **100** appropriately performs bone detection using a certain technology, which may be any technology as long as bone detection can be performed using an image, and a detailed description thereof will be omitted. Subsequently, the information processing apparatus **100** performs context estimation using bone information indicating a detection result of the bone detection (step S502). The information processing apparatus **100** appropriately performs context estimation using a certain technology, which may be any technology as long as context estimation can be performed using the detection result of bone detection, and detailed description will be omitted.

[0224] Subsequently, the information processing apparatus **100** judges whether a specific context has been detected (step S503). In a case where the specific context has not been detected (step S503: No), the information processing apparatus **100** returns to step S501 and repeats the processing.

[0225] In a case where the specific context has been detected (step S503: Yes), the information

processing apparatus **100** calculates a motion vector of the feature point (step S504). Subsequently, the information processing apparatus **100** calculates an intersection between each motion vector and the module upper surface (step S505).

[0226] The information processing apparatus **100** judges whether there is a feature point whose distance to the module upper surface is within a threshold (step S506). In a case where there is no feature point whose distance to the module upper surface is within the threshold (step S506: No), the information processing apparatus **100** returns to step S501 and repeats the processing. In a case where there is a feature point whose distance to the module upper surface is within the threshold (step S506: Yes), the information processing apparatus **100** specifies a module to be deformed (step S507).

[0227] Subsequently, the information processing apparatus **100** determines softness (step S508). The information processing apparatus **100** drives the motor (step S509). For example, the information processing apparatus **100** controls the softness/firmness controller **30** according to the determined firmness of the firmness alterable module **10**. Subsequently, the information processing apparatus **100** returns to step S501 and repeats the processing.

[0228] As described above, the information processing apparatus **100** executes firmness control by context based on sensing by a camera.

[0229] The information processing apparatus **100** performs bone detection. The information processing apparatus **100** estimates the context from the bone information. In a case where a specific context has been detected, the information processing apparatus **100** calculates a motion vector of each feature point of the object. For example, in an exemplary case of a desk, the information processing apparatus **100** calculates a motion vector of each feature point of the object in a case where a specific context such as resting an elbow or facing down has been detected.

[0230] The information processing apparatus **100** calculates an intersection between the motion vector and the module upper surface. In a case where there is a feature point whose distance to the module upper surface is within a threshold, the information processing apparatus **100** defines a module including an intersection of a motion vector of the feature point and the module upper surface, as a target module to undergo deformation. The information processing apparatus **100** determines softness. The information processing apparatus **100** may prepare a library of softness for the target object in advance and refer to the value. In a case where a pressure sensor is included in the module, the information processing apparatus **100** may calculate softness with respect to pressure.

[0231] The information processing apparatus **100** may alter the softness in consideration of the moving speed of the feature point. When the module to be deformed is determined, the information processing apparatus **100** may drive the motor to set the module in a soft state. In a case where a module that is already soft is excluded from the target, the information processing apparatus **100** may return the module to the firm state. In a case where there is no feature point whose distance to the module upper surface is within the threshold, the information processing apparatus **100** may drive the motor so that all the modules are in firm states.

1-5-5. Processing Example According to User

[0232] The information processing apparatus **100** may alter the firmness of the module according to the user. For example, the information processing apparatus **100** may individually identify the user of the object **2** and alter the firmness of the object **2** to firmness according to the user. For example, the information processing apparatus **100** may alter the firmness of the object **2** such as a car seat, an office chair, or a bed to personalized firmness.

[0233] First, an example of a flow of processing according to a user will be described with reference to FIG. 22. FIG. 22 is a flowchart illustrating a procedure of processing according to object detection. Note that description of points similar to the description in FIGS. 14, 18, and the like will be omitted as appropriate.

[0234] As illustrated in FIG. 22, in a case where the switch is not turned ON (step S601: No), the

information processing apparatus **100** drives the actuator until a preset state is achieved (step **S602**). For example, in a case where the switch for firmness alteration is OFF and the mode is not the firmness alteration mode, the information processing apparatus **100** controls the softness/firmness controller **30** until all the firmness alterable modules **10** of the object **2** as the target object are in preset states (default state). Subsequently, the information processing apparatus **100** maintains the state set in step **S602** and stops firmness alteration.

[0235] When the switch is turned ON (step **S601**: Yes), the information processing apparatus **100** performs image recognition (step **S603**). The information processing apparatus **100** appropriately performs image recognition using a certain technology, which may be any technology as long as image recognition for specifying the user can be performed, and a detailed description thereof will be omitted. For example, in a case where the switch for firmness alteration is ON and the mode is the firmness alteration mode, the information processing apparatus **100** performs image recognition using an image captured by the sensor apparatus **50** being a camera.

[0236] Subsequently, the information processing apparatus **100** judges whether a registered person has been detected (step **S604**). In a case where the registered person has not been detected (step **S604**: No), the information processing apparatus **100** returns to step **S601** and repeats the processing.

[0237] In a case where the registered person has been detected (step **S604**: Yes), the information processing apparatus **100** determines softness (step **S605**). For example, in a case where the registered person has been detected, the information processing apparatus **100** acquires setting information of the firmness of the object **2** corresponding to the person from the storage unit **120**, and determines the firmness of each of the firmness alterable modules **10** using the acquired setting information of the firmness of the object **2**.

[0238] Subsequently, the information processing apparatus **100** drives the actuator (step **S606**). For example, the information processing apparatus **100** controls the softness/firmness controller **30** according to the determined firmness of the firmness alterable module **10**. Subsequently, the information processing apparatus **100** returns to step **S601** and repeats the processing.

[0239] As described above, in an interaction case corresponding to the user, the information processing apparatus **100** executes various types of processing according to ON and OFF of the switch. For example, the information processing apparatus **100** performs image recognition in a case where the switch is ON. Subsequently, in a case where the registered person is detected, the information processing apparatus **100** acquires softness (deformation amount) information of each module from the library, and drives the actuator using the acquired information.

[0240] In a case where the switch is OFF, the information processing apparatus **100** drives the actuator so as to set all the modules in preset states. Furthermore, the information processing apparatus **100** recognizes (specifies) the user included in the image by using a certain personal identification method. The information processing apparatus **100** may use not only image recognition but also information of a tag, a smartphone, and biological information. Furthermore, the information processing apparatus **100** may use personalized firmness (deformation amount) information. For example, the information processing apparatus **100** may use preregistered information. For example, the information processing apparatus **100** may update the information by acquiring habits of the user at the time of use from the pressure distribution or the posture information by the camera and performing machine learning using the data of habits.

[0241] Note that the above-described change in firmness according to the user is merely an example, and the information processing apparatus **100** may alter the firmness by appropriately using various information related to the user. For example, in a case where there is a part that should not be pressed due to injury or the like, the information processing apparatus **100** may alter the firmness distribution of the surface of the object **2** being a bed so as not to apply pressure to the affected part. First, an outline of an interaction case according to a physical part of the user will be described with reference to FIG. **23**. FIG. **23** is a conceptual diagram illustrating an example of

alteration according to a physical part of the user.

[0242] FIG. 23 illustrates a state in which a user U having a physical part AP in the abdomen as an affected part is lying on the object 2 being a bed. The information processing apparatus 100 specifies the position on the object 2 where the physical part AP of the user U is located by using the image obtained by imaging the object 2 being bed on which the user U is lying and the physical part AP of the user U. Subsequently, the information processing apparatus 100 determines to soften, that is, decrease the firmness of the module disposed at the position on the object 2 where the physical part AP of the user U is located. In FIG. 23, the information processing apparatus 100 determines to soften, that is, decrease the firmness of the hatched module, being the fourth module from the left.

[0243] In this manner, in a case where there is an affected part such as the case of the user U, the information processing apparatus 100 alters the firmness of the position on the object 2 corresponding to the affected part of the user U. For example, in a case where a person with an affected part is lying on the object 2, the information processing apparatus 100 alters the firmness of the module corresponding to the surface of the object 2 so that a place where the affected part is located becomes a soft surface.

[0244] Hereinafter, an example of a flow of processing according to the physical part of the user will be described with reference to FIG. 24. FIG. 24 is a flowchart illustrating a procedure of processing according to a physical part of a user. Note that description of points similar to the description in FIGS. 14, 21, 22, and the like will be omitted as appropriate.

[0245] As illustrated in FIG. 24, in a case where the switch is not turned ON (step S701: No), the information processing apparatus 100 drives the actuator until each module becomes a preset state (step S702). For example, in a case where the switch for firmness alteration is OFF and the mode is not the firmness alteration mode, the information processing apparatus 100 controls the softness/firmness controller 30 until all the firmness alterable modules 10 of the object 2 as the target object are in preset states (default state). Subsequently, the information processing apparatus 100 maintains the state set in step S702 and stops firmness alteration.

[0246] In a case where the switch is ON (step S701: Yes), the information processing apparatus 100 performs bone detection (step S703). Subsequently, the information processing apparatus 100 estimates the position of the affected part using the bone information indicating the detection result of the bone detection (step S704). The information processing apparatus 100 appropriately performs position estimation of the affected part using a certain technology, which may be any technology as long as position estimation of the affected part can be performed using the detection result of bone detection, and detailed description will be omitted.

[0247] Subsequently, the information processing apparatus 100 specifies a module to be transformed (step S705). For example, the information processing apparatus 100 determines the firmness alterable module 10 corresponding to position of the affected part in the object 2 as a target module to undergo firmness alteration.

[0248] Subsequently, the information processing apparatus 100 determines softness (step S706). For example, the information processing apparatus 100 determines the softness of the firmness alterable module 10 determined as the target module. For example, the information processing apparatus 100 determines to decrease the firmness of the firmness alterable module 10 such that the larger the affected part position, the more the firmness of the firmness alterable module 10 is to be decreased.

[0249] Subsequently, the information processing apparatus 100 drives the actuator (step S707). For example, the information processing apparatus 100 controls the softness/firmness controller 30 according to the determined firmness of the firmness alterable module 10. Subsequently, the information processing apparatus 100 returns to step S701 and repeats the processing.

[0250] As described above, in an interaction case corresponding to the physical part of the user, the information processing apparatus 100 executes various types of processing according to ON and

OFF of the switch. For example, in a case where the switch is ON, the information processing apparatus **100** executes the following processing.

[0251] The information processing apparatus **100** performs object estimation. The information processing apparatus **100** detects a bone. The information processing apparatus **100** estimates the affected part position on the bone based on information of the affected part position and the bone information. Based on the affected part position information, the information processing apparatus **100** sets a module near the affected part position as a target module to undergo deformation. The information processing apparatus **100** determines softness. The information processing apparatus **100** may use firmness (deformation amount) set in advance. When the module to be deformed is determined, the information processing apparatus **100** may drive the actuator to set the module in a soft state. Furthermore, in a case where a module that is already soft is excluded from the target, the information processing apparatus **100** may return the module to the preset state.

[0252] In a case where the switch is OFF, the information processing apparatus **100** drives the actuator so as to set all the modules in preset states.

[1-5-6. Processing Example According to Force Applied to User]

[0253] Furthermore, the information processing apparatus **100** may alter the firmness of the module according to the force applied to the user. For example, the information processing apparatus **100** may dynamically alter the firmness according to the change in inertia accompanying movement. For example, the information processing apparatus **100** alters the firmness of a part of a seat, which is an example of the object **2** and is normally soft, so as to support the body of the user when the body of the user is pressed against the seat due to an inertial force when turning a curve on a road. First, an outline of another interaction case by sensing according to the force applied to the user will be described with reference to FIG. **25**. FIG. **25** is a conceptual diagram illustrating an example of alteration according to a force applied to a user.

[0254] FIG. **25** illustrates a state in which a user **U** is driving while sitting on an object **2** being a seat of an automobile. In FIG. **25**, the object **2** before the firmness alteration is described as an “object **2-21**”, and the object **2** after the firmness alteration is described as an “object **2-22**”. However, the object **2** will be described as the object **2** when described with no particular distinction from each other. The object **2-21** indicates a state in which the lateral force applied to the user **U** is less than a predetermined value. In addition, the object **2-22** indicates a state in which the lateral force applied to the user **U** is a predetermined value or more.

[0255] In a case where the lateral force applied to the user **U** is the predetermined value or more, the information processing apparatus **100** determines to alter the firmness of the module (step **S21**). In a case where the lateral force applied to the user **U** alters from the state indicated by the object **2-21** to the state indicated by the object **2-22**, the information processing apparatus **100** determines to alter the firmness of the module located in the moving direction of the body of the user **U**. In FIG. **25**, the information processing apparatus **100** determines to alter the firmness of the module of the hatched portion on the right side of the seat. In this case, the firmness of the module of the hatched portion may be increased or decreased on the right side of the seat of the object **2**. For example, the information processing apparatus **100** may determine to increase the firmness to make the module firm in the hatched portion on the right side of the seat. For example, the information processing apparatus **100** may determine to soften, that is, decrease the firmness of the module in the hatched portion on the right side of the seat.

[0256] In this manner, the information processing apparatus **100** alters the firmness when the force applied to the user **U** satisfies a predetermined condition. As described above, the information processing apparatus **100** alters the firmness of the module in a case where the inertial force has greatly changed.

[0257] Hereinafter, an example of a flow of processing according to object detection will be described with reference to FIG. **26**. FIG. **26** is a flowchart illustrating a procedure of processing according to a force applied to a user. Note that description of points similar to the description in

FIGS. 14, 18, 22, and the like will be omitted as appropriate.

[0258] As illustrated in FIG. 26, in a case where the switch is not turned ON (step S801: No), the information processing apparatus 100 drives the actuator until the preset state is achieved (step S802). Subsequently, the information processing apparatus 100 maintains the state set in step S802 and stops firmness alteration.

[0259] When the switch is turned ON (step S801: Yes), the information processing apparatus 100 acquires the steering wheel angle and speed information (step S803). The information processing apparatus 100 acquires information indicating the angle of the steering wheel and speed information indicating the speed detected by the sensor apparatus 50 such as an accuracy sensor and a speed sensor. For example, in a case where the switch for firmness alteration is ON and the mode is the firmness alteration mode, the information processing apparatus 100 acquires steering wheel angle information and speed information detected by the sensor apparatus 50.

[0260] Subsequently, the information processing apparatus 100 calculates an inertial force applied to the user (step S804). For example, the information processing apparatus 100 calculates the inertial force applied to the user by using the steering wheel angle information and the speed information.

[0261] The information processing apparatus 100 judges whether the inertial force is a threshold or more (step S805). For example, the information processing apparatus 100 judges whether the inertial force applied to the user is the threshold or more.

[0262] In a case where the inertial force is not the threshold or more (step S805: No), the information processing apparatus 100 returns to step S801 and repeats the processing. That is, in a case where the inertial force applied to the user is less than the threshold, the information processing apparatus 100 returns to step S801 and repeats the processing without altering the firmness.

[0263] In a case where the inertial force is the threshold or more (step S805: Yes), the information processing apparatus 100 specifies a module to be deformed (step S806). For example, in a case where the inertial force is the threshold or more, the information processing apparatus 100 determines the firmness alterable module 10 located in an acting direction of the inertial force among the objects 2 as a target module to undergo firmness alteration.

[0264] Subsequently, the information processing apparatus 100 determines softness (step S807). For example, the information processing apparatus 100 determines the softness of the firmness alterable module 10 determined as the target module. The information processing apparatus 100 determines to decrease the firmness of the firmness alterable module 10 such that the greater the inertial force on the user, the more the firmness of the firmness alterable module 10 being a target module is to be decreased.

[0265] Subsequently, the information processing apparatus 100 drives the actuator (step S808). For example, the information processing apparatus 100 controls the softness/firmness controller 30 according to the determined firmness of the firmness alterable module 10. Subsequently, the information processing apparatus 100 returns to step S801 and repeats the processing.

[0266] As described above, in an interaction case using camera sensing, the information processing apparatus 100 executes various types of processing according to ON and OFF states of the switch. For example, in a case where the switch is ON, the information processing apparatus 100 executes the following processing.

[0267] The information processing apparatus 100 acquires steering wheel angle, speed information, and the like. The information processing apparatus 100 calculates an inertial force applied to the user based on the acquired information. In a case where the inertial force is a threshold or more, the information processing apparatus 100 specifies a module to be deformed based on the direction of the inertial force. The information processing apparatus 100 determines softness. The information processing apparatus 100 may perform calculation based on the current (pre-change) firmness and the value of the inertial force. The information processing apparatus 100 drives an actuator.

[0268] In a case where the switch is OFF, the information processing apparatus **100** drives the actuator so as to set all the modules in preset states.

[1-5-7. Other Information Examples]

[0269] Note that the above-described processing is merely an example, and the information processing apparatus **100** may perform processing by appropriately using various types of information. For example, the information processing apparatus **100** may perform an interaction using information detected by various sensors, not limited to the pressure sensor or the camera, as an implicit input.

[0270] For example, the information processing apparatus **100** may use biological information of the user. For example, the information processing apparatus **100** may use information related to respiration of the user. For example, the information processing apparatus **100** may alter the firmness of the object **2** for the purpose of relaxation so as to control the breathing rhythm of the user with the object **2**, which is a cushion that expands or contracts, as a target object. For example, the information processing apparatus **100** may alter the firmness of the object **2** so as to bring about a relaxation effect on the user by controlling the respiration of the user by alternately switching the partial softness/firmness of the backrest of the object **2**.

[0271] For example, the information processing apparatus **100** may use the electroencephalogram of the user. For example, the information processing apparatus **100** may predict the action of the user from the electroencephalogram pattern to alter the firmness of the object **2** being the target object.

[0272] For example, the information processing apparatus **100** may use verbal input by the user. For example, the information processing apparatus **100** may alter the firmness of the object **2** such that, when a voice command of the user is recognized by the voice recognition technology, the firmness is altered to the instructed firmness.

[0273] For example, the information processing apparatus **100** may use information regarding a change in the external environment. For example, in a case where an earthquake occurs, the information processing apparatus **100** may judge whether another object (such as a vase) being a fragile object is placed on the object **2** from the information obtained by object recognition. When having judged that another object being a fragile object is placed on the object **2**, the information processing apparatus **100** may alter the firmness of the object **2** so as to decrease the firmness of the periphery of the position of the object **2** on which the other object is placed in order to prevent breakage even if the object **2** falls down.

2. OTHER EMBODIMENTS

[0274] The process according to each of embodiments described above may be performed in various different forms (modifications) in addition to each of embodiments or modifications described above.

2-1. Other Configuration Examples

[0275] Although the above information processing system **1** has been described using the example in which the information processing apparatus **100** is disposed in the object **2**. However, the above configuration is an example, and as long as a service related to the firmness alteration of the surface of the object **2** can be provided, the information processing system **1** can arrange the apparatus in any manner. For example, the information processing system **1** may have a configuration in which the information processing apparatus **100** is located outside the object **2**, and the information processing apparatus **100** communicates with the firmness alterable module **10** located in the object **2**, so as to allow the information processing apparatus **100** to control the firmness of the firmness alterable module **10**.

2-2. Others

[0276] Furthermore, among each processing described in the above embodiments, all or a part of the processing described as being performed automatically can be manually performed, or the processing described as being performed manually can be performed automatically by a known

method. In addition, the processing procedures, specific names, and information including various data and parameters depicted in the above specifications or drawings can be altered in any manner unless otherwise specified. For example, various types of information illustrated in each of the drawings are not limited to the information illustrated.

[0277] In addition, each of components of each device is provided as a functional and conceptional illustration and thus does not necessarily need to be physically configured as illustrated. That is, the specific form of distribution/integration of each device is not limited to those depicted in the drawings, and all or a part thereof may be functionally or physically distributed or integrated into arbitrary units according to various loads and use conditions.

[0278] Furthermore, the above-described embodiments and modifications can be appropriately combined within a range implementable without contradiction of processing.

[0279] The effects described in the present specification are merely examples, and thus, there may be other effects, not limited to the exemplified effects.

3. EFFECTS ACCORDING TO PRESENT DISCLOSURE

[0280] As described above, the information processing apparatus (the information processing apparatus **100** in the embodiment) according to the present disclosure includes the acquisition unit (the acquisition unit **131** in the embodiment), the estimation unit (the estimation unit **132** in the embodiment), and the determination unit (the determination unit **133** in the embodiment). The acquisition unit acquires sensor information detected by the sensor. Based on the sensor information acquired by the acquisition unit, the estimation unit estimates a contact mode of an object with respect to the target whose surface firmness is alterable. The determination unit determines whether to alter the firmness of the surface of the target according to the contact mode of the object estimated by the estimation unit.

[0281] In this manner, by determining whether to alter the firmness of the surface of the target according to the contact mode of the object with respect to the target object estimated based on the sensor information detected by the sensor, the information processing apparatus according to the present disclosure can alter the firmness of the surface of the target with no explicit instruction from the user. Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0282] The acquisition unit acquires sensor information including pressure information indicating the pressure on the surface of the target detected by the pressure sensor. The estimation unit estimates a contact mode of the object with the surface of the target based on the pressure information. In this manner, by determining whether to alter the firmness of the surface of the target according to the pressure to the surface of the target, the information processing apparatus according to the present disclosure can alter the firmness of the surface of the target with no explicit instruction from the user. Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0283] The estimation unit calculates a change amount in the pressure applied by the object to the surface of the target. The determination unit determines whether to alter the firmness of the surface of the target based on the change amount in the pressure. In this manner, by determining whether to alter the firmness of the surface of the target according to the change amount of pressure to the surface of the target, the information processing apparatus according to the present disclosure can alter the firmness of the surface of the target with no explicit instruction from the user.

Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0284] In a case where the change amount in the pressure is a threshold or more, the determination unit determines to alter the firmness of the surface of the target. In this manner, by determining to alter the firmness of the surface of the target when the change amount of pressure to the surface of the target is a threshold or more, the information processing apparatus according to the present disclosure can alter the firmness of the surface of the target with no explicit instruction from the

user. Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0285] In a case where a gesture corresponding to a pressure change pattern satisfies a condition, the determination unit determines to alter the firmness of the surface of the target. In this manner, by determining to alter the firmness of the surface of the target in the case where the gesture corresponding to the pressure change pattern satisfies the condition, the information processing apparatus according to the present disclosure can alter the firmness of the surface of the target with no explicit instruction from the user. Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0286] The determination unit determines the alteration amount in the firmness of the surface of the target according to the change amount in the pressure. In this manner, by determining the alteration amount in the firmness of the surface of the target according to the change amount of pressure, the information processing apparatus according to the present disclosure can alter the firmness of the surface of the target with no explicit instruction from the user. Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0287] The acquisition unit acquires sensor information including pressure information indicating pressure to each of the plurality of regions on the surface of the target. Based on the pressure information, the estimation unit calculates a change amount of the pressure applied by the object to each of the plurality of regions on the surface of the target. The determination unit determines a region in which the change amount in the pressure satisfies the condition among the plurality of regions on the surface of the target, as the region to undergo firmness alteration. In this manner, by determining the region where the change amount in the pressure satisfies the condition among the plurality of regions on the surface of the target, as a region to undergo firmness alteration, the information processing apparatus can alter the partial firmness of the surface of the target.

Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0288] The acquisition unit acquires sensor information including image information captured by the image sensor. Based on the image information, the estimation unit estimates a contact mode of the object with the surface of the target. In this manner, by estimating a contact mode of the object with the surface of the target based on the image information and determining whether to alter the firmness of the surface of the target according to the estimated contact mode of the object with the target object, the information processing apparatus according to the present disclosure can alter the firmness of the surface of the target with no explicit instruction from the user. Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0289] The estimation unit calculates an intersection between a feature point of the object and the surface of the target by using a moving direction of the object based on the image information. In a case where the distance between the feature point of the object and the intersection is the threshold or more, the determination unit determines to alter the firmness of the surface of the target. In this manner, by determining to alter the firmness of the surface of the target when the distance between the feature point of the object and the intersection is a threshold or more, the information processing apparatus according to the present disclosure can alter the firmness of the surface of the target with no explicit instruction from the user. Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0290] The determination unit determines the alteration amount in the firmness of the surface of the target according to the moving speed of the object. In this manner, by determining the alteration amount in the firmness of the surface of the target according to the moving speed of the object, the information processing apparatus according to the present disclosure can alter the firmness of the surface of the target with no explicit instruction from the user. Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0291] Furthermore, the estimation unit estimates the context related to the object based on the

image information. When the context estimated by the estimation unit satisfies a condition, the determination unit determines to alter the firmness of the surface of the target. In this manner, by determining to alter the firmness of the surface of the target when the estimated context satisfies the condition, the information processing apparatus according to the present disclosure can alter the firmness of the surface of the target with no explicit instruction from the user. Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0292] The determination unit determines the alteration amount in the firmness of the surface of the target according to the context. In this manner, by determining the alteration amount in the firmness of the surface of the target according to the context, the information processing apparatus according to the present disclosure can alter the firmness of the surface of the target with no explicit instruction from the user. Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0293] The acquisition unit acquires sensor information including image information obtained by imaging the user by the image sensor. The estimation unit estimates information related to the user based on the image information. The determination unit determines whether to alter the firmness of the surface of the target based on the information related to the user estimated by the estimation unit. In this manner, by determining whether to alter the firmness of the surface of the target based on the information related to the user which is estimated based on the image information including the image of the user, the information processing apparatus according to the present disclosure can alter the firmness of the surface of the target with no explicit instruction from the user. Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0294] The estimation unit estimates a contact mode of the object with respect to a plurality of regions on the surface of the target based on the image information. The determination unit determines a region in which the contact mode satisfies the condition among the plurality of regions on the surface of the target, as a region to undergo firmness alteration. In this manner, by determining the region in which the contact mode satisfies the condition among the plurality of regions on the surface of the target, as a region to undergo firmness alteration, the information processing apparatus can alter the partial firmness of the surface of the target. Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0295] As described above, the firmness alterable module (the firmness alterable module **10** in the embodiment) according to the present disclosure is a firmness alterable module constituting at least a part of a surface of a target, and includes a coupling portion (the coupling portion **21** in the embodiment), a block portion (the block portion **22** in the embodiment), and a softness/firmness controller (the softness/firmness controller **30** in the embodiment). The coupling portion is a sheet-like member disposed along the surface and flexible in a direction intersecting the surface. The block portion includes a plurality of blocks (the blocks **220** in the embodiment) aligned along a back surface which is a surface on a side opposite to the surface of the coupling portion and coupled to the back surface. The softness/firmness controller is disposed along one surface of each of the plurality of blocks with the block portion interposed between the softness/firmness controller and the coupling portion, and can alter firmness.

[0296] In this manner, the firmness alterable module according to the present disclosure includes: the sheet-like coupling portion disposed along the surface and flexible in the direction intersecting the surface; and the plurality of blocks disposed side by side along the back surface on the opposite side of the surface of the coupling portion and coupled to the back surface, making it possible to suppress the change in the shape of the surface due to the pressure toward the surface side. Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0297] In addition, each of the plurality of blocks is a hexahedron in which a facing surface facing one surface is coupled to the coupling portion, while surfaces other than the one surface and the

facing surface are disposed adjacent to surfaces of other blocks. In this manner, the firmness alterable module has each of the plurality of blocks disposed adjacent to each other. This allows the blocks to interfere with each other when the pressure to the surface side is generated, making it possible to suppress the change in the shape of the surface due to the pressure to the surface side. Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0298] In addition, the softness/firmness controller includes a flexible member (the flexible member **31** in the embodiment) and a pressing mechanism (the pressing mechanism **32** in the embodiment). The flexible member is disposed along one surface of each of the plurality of blocks. The pressing mechanism presses the flexible member toward the block portion from a position where the block portion and the flexible member are sandwiched. In this manner, since the firmness alterable module has the flexible member, the firmness of the surface of the target can be altered by pressing the flexible member. Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0299] In addition, the pressing mechanism includes a movable portion (the movable portion **321** in the embodiment) and a drive unit (the drive unit **322** in the embodiment). The movable portion has a plate shape along the flexible member at a position sandwiching the block portion and the flexible member. The drive unit moves the movable portion in a direction intersecting the surface. In this manner, the firmness alterable module can change the firmness of the surface of the target by pressing the flexible member with the plate-shaped movable portion. Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

[0300] In addition, the firmness alterable module includes a cover portion (the cover portion **40** in the embodiment). The cover portion covers the coupling portion, the block portion, and the softness/firmness controller, and an object comes in contact with a surface opposite to a surface along the coupling portion. In this manner, the firmness alterable module has a member that covers the outer surface of the firmness alterable module and comes into contact with an external object, making it possible to appropriately hold the shape of the surface. Consequently, the information processing apparatus can appropriately alter the firmness of the surface of the object.

4. HARDWARE CONFIGURATION

[0301] The information apparatus such as the information processing apparatus **100** according to the above-described individual embodiments and modifications are implemented by a computer **1000** having a configuration as illustrated in FIG. 27, for example. FIG. 27 is a hardware configuration diagram illustrating an example of the computer **1000** that implements the functions of the information processing apparatus such as the information processing apparatus **100**.

Hereinafter, the information processing apparatus **100** according to the embodiment will be described as an example. The computer **1000** includes a CPU **1100**, RAM **1200**, read only memory (ROM) **1300**, a hard disk drive (HDD) **1400**, a communication interface **1500**, and an input/output interface **1600**. Individual components of the computer **1000** are interconnected by a bus **1050**.

[0302] The CPU **1100** operates based on a program stored in the ROM **1300** or the HDD **1400** so as to control each of components. For example, the CPU **1100** develops the program stored in the ROM **1300** or the HDD **1400** into the RAM **1200** and executes processing corresponding to various programs.

[0303] The ROM **1300** stores a boot program such as a basic input output system (BIOS) executed by the CPU **1100** when the computer **1000** starts up, a program dependent on hardware of the computer **1000**, or the like.

[0304] The HDD **1400** is a non-transitory computer-readable recording medium that records a program executed by the CPU **1100**, data used by the program, or the like. Specifically, the HDD **1400** is a recording medium that records an information processing program according to the present disclosure, which is an example of program data **1450**.

[0305] The communication interface **1500** is an interface for connecting the computer **1000** to an

external network **1550** (for example, the Internet). For example, the CPU **1100** receives data from other devices or transmits data generated by the CPU **1100** to other devices via the communication interface **1500**.

[0306] The input/output interface **1600** is an interface for connecting an input/output device **1650** with the computer **1000**. For example, the CPU **1100** receives data from an input device such as a keyboard or a mouse via the input/output interface **1600**. In addition, the CPU **1100** transmits data to an output device such as a display, a speaker, or a printer via the input/output interface **1600**. Furthermore, the input/output interface **1600** may function as a media interface for reading a program or the like recorded on predetermined recording medium (or simply medium). Examples of the media include optical recording media such as a digital versatile disc (DVD) or a phase change rewritable disk (PD), a magneto-optical recording medium such as a magneto-optical disk (MO), a tape medium, a magnetic recording medium, and semiconductor memory.

[0307] For example, when the computer **1000** functions as the information processing apparatus **100** according to the embodiment, the CPU **1100** of the computer **1000** executes the information processing program loaded on the RAM **1200** so as to implement the functions of the control unit **130** and the like. Furthermore, the HDD **1400** stores the information processing program according to the present disclosure or data in the storage unit **120**. While the CPU **1100** executes program data **1450** read from the HDD **1400**, the CPU **1100** may acquire these programs from another device via the external network **1550**, as another example.

[0308] Note that the present technology can also have the following configurations.

(1)

[0309] An information processing apparatus comprising: [0310] an acquisition unit that acquires sensor information detected by a sensor; [0311] an estimation unit that estimates, based on the sensor information acquired by the acquisition unit, a contact mode of an object with respect to a target of which a firmness of a surface is alterable; and [0312] a determination unit that determines whether to alter the firmness of the surface of the target according to the contact mode of the object estimated by the estimation unit.

(2)

[0313] The information processing apparatus according to (1), wherein [0314] the acquisition unit [0315] acquires the sensor information including pressure information indicating a pressure applied to the surface of the target detected by a pressure sensor, and [0316] the estimation unit [0317] estimates the contact mode of the object with the surface of the target based on the pressure information.

(3)

[0318] The information processing apparatus according to (2), wherein [0319] the estimation unit [0320] calculates a change amount in the pressure applied by the object to the surface of the target, and [0321] the determination unit [0322] determines whether to alter the firmness of the surface of the target based on the change amount in the pressure.

(4)

[0323] The information processing apparatus according to (3), wherein [0324] the determination unit [0325] determines to alter the firmness of the surface of the target in a case where the change amount in the pressure is a threshold or more.

(5)

[0326] The information processing apparatus according to (3), wherein [0327] the determination unit [0328] determines to alter the firmness of the surface of the target in a case where a gesture corresponding to a pressure change pattern satisfies a condition.

(6)

[0329] The information processing apparatus according to any one of (3) to (5), wherein [0330] the determination unit [0331] determines an alteration amount in the firmness of the surface of the target according to the change amount in the pressure.

(7)
[0332] The information processing apparatus according to any one of (3) to (6), wherein [0333] the acquisition unit [0334] acquires the sensor information including the pressure information indicating a pressure to each of a plurality of regions on the surface of the target, [0335] the estimation unit [0336] calculates, based on the pressure information, the change amount in the pressure applied by the object to each of the plurality of regions on the surface of the target, and [0337] the determination unit [0338] determines a region in which the change amount in the pressure satisfies a condition, among the plurality of regions on the surface of the target, as a region to undergo firmness alteration.

(8)
[0339] The information processing apparatus according to any one of (1) to (7), wherein [0340] the acquisition unit [0341] acquires the sensor information including image information imaged by an image sensor, and [0342] the estimation unit [0343] estimates the contact mode of the object with the surface of the target based on the image information.

(9)
[0344] The information processing apparatus according to (8), wherein [0345] the estimation unit [0346] calculates an intersection between a feature point of the object and the surface of the target by using a moving direction of the object based on the image information, and [0347] the determination unit [0348] determines to alter the firmness of the surface of the target in a case where a distance between the feature point of the object and the intersection is a threshold or more.

(10)
[0349] The information processing apparatus according to (9), wherein [0350] the determination unit [0351] determines an alteration amount in the firmness of the surface of the target according to a moving speed of the object.

(11)
[0352] The information processing apparatus according to (9) or (10), wherein [0353] the estimation unit [0354] estimates a context related to the object based on the image information, and [0355] the determination unit [0356] determines to alter the firmness of the surface of the target in a case where the context estimated by the estimation unit satisfies a condition.

(12)
[0357] The information processing apparatus according to (11), wherein [0358] the determination unit [0359] determines an alteration amount in the firmness of the surface of the target according to the context.

(13)
[0360] The information processing apparatus according to any one of (8) to (12), wherein [0361] the acquisition unit [0362] acquires the sensor information including the image information obtained by imaging a user by the image sensor, [0363] the estimation unit [0364] estimates information related to the user based on the image information, and [0365] the determination unit [0366] determines whether to alter the firmness of the surface of the target based on the information related to the user estimated by the estimation unit.

(14)
[0367] The information processing apparatus according to any one of (8) to (13), wherein [0368] the estimation unit [0369] estimates the contact mode of the object with respect to a plurality of regions on the surface of the target based on the image information, and [0370] the determination unit [0371] determines a region in which the contact mode satisfies a condition among the plurality of regions on the surface of the target, as a region to undergo firmness alteration.

(15)
[0372] An information processing method comprising: [0373] acquiring sensor information detected by a sensor; [0374] estimating, based on the acquired sensor information, a contact mode of an object with respect to a target of which a firmness of a surface is alterable; and [0375]

determining whether to alter the firmness of the surface of the target according to the estimated contact mode of the object.

(16)

[0376] A firmness alterable module constituting at least a part of a surface of a target, the firmness alterable module comprising: [0377] a sheet-like coupling portion that is disposed along the surface and is flexible in a direction intersecting the surface; [0378] a block portion having a plurality of blocks disposed side by side along a back surface and coupled to the back surface, the back surface being a surface opposite to the surface of the coupling portion; and [0379] a softness/firmness controller that is disposed along one surface of each of the plurality of blocks with the block portion interposed between the softness/firmness controller and the coupling portion, the softness/firmness controller being capable of altering firmness.

(17)

[0380] The firmness alterable module according to (16), wherein [0381] each of the plurality of blocks is [0382] a hexahedron in which a facing surface facing the one surface is coupled to the coupling portion, and surfaces other than the one surface and the facing surface are disposed adjacent to surfaces of other blocks.

(18)

[0383] The firmness alterable module according to (16) or (17), wherein [0384] the softness/firmness controller includes: [0385] a flexible member disposed along one surface of each of the plurality of blocks; and [0386] a pressing mechanism that presses the flexible member toward the block portion from a position where the block portion and the flexible member are sandwiched.

(19)

[0387] The firmness alterable module according to (18), wherein [0388] the pressing mechanism includes: [0389] a movable portion having a plate shape along the flexible member at a position sandwiching the block portion and the flexible member; and [0390] a drive unit that moves the movable portion in a direction intersecting the surface.

(20)

[0391] The firmness alterable module according to any one of (16) to (19), further comprising [0392] a cover portion that covers the coupling portion, the block portion, and the softness/firmness controller, and to which an object comes in contact with a surface being an opposite surface of a surface running along the coupling portion.

REFERENCE SIGNS LIST

[0393] **1** INFORMATION PROCESSING SYSTEM [0394] **100** INFORMATION PROCESSING APPARATUS [0395] **110** COMMUNICATION UNIT [0396] **120** STORAGE UNIT [0397] **121** MODULE INFORMATION STORAGE UNIT [0398] **122** HISTORY INFORMATION STORAGE UNIT [0399] **123** SENSOR INFORMATION STORAGE UNIT [0400] **124** ALTERATION CONDITION INFORMATION STORAGE UNIT [0401] **125** USER INFORMATION STORAGE UNIT [0402] **130** CONTROL UNIT [0403] **131** ACQUISITION UNIT [0404] **132** ESTIMATION UNIT [0405] **133** DETERMINATION UNIT [0406] **134** PROCESSING UNIT [0407] **135** TRANSMISSION UNIT [0408] **10** FIRMNESS ALTERABLE MODULE [0409] **20** SURFACE FORMING MEMBER [0410] **21** COUPLING PORTION [0411] **22** BLOCK PORTION [0412] **220** BLOCK [0413] **221** FIRST SURFACE [0414] **222** SECOND SURFACE [0415] **30** SOFTNESS/FIRMNESS CONTROLLER [0416] **31** FLEXIBLE MEMBER [0417] **32** PRESSING MECHANISM [0418] **321** MOVABLE PORTION [0419] **322** DRIVE UNIT [0420] **323** SHAFT UNIT [0421] **324** SUPPORT UNIT [0422] **40** COVER PORTION [0423] **50** SENSOR APPARATUS

Claims

1. An information processing apparatus comprising: an acquisition unit that acquires sensor information detected by a sensor; an estimation unit that estimates, based on the sensor information acquired by the acquisition unit, a contact mode of an object with respect to a target of which a firmness of a surface is alterable; and a determination unit that determines whether to alter the firmness of the surface of the target according to the contact mode of the object estimated by the estimation unit.
2. The information processing apparatus according to claim 1, wherein the acquisition unit acquires the sensor information including pressure information indicating a pressure applied to the surface of the target detected by a pressure sensor, and the estimation unit estimates the contact mode of the object with the surface of the target based on the pressure information.
3. The information processing apparatus according to claim 2, wherein the estimation unit calculates a change amount in the pressure applied by the object to the surface of the target, and the determination unit determines whether to alter the firmness of the surface of the target based on the change amount in the pressure.
4. The information processing apparatus according to claim 3, wherein the determination unit determines to alter the firmness of the surface of the target in a case where the change amount in the pressure is a threshold or more.
5. The information processing apparatus according to claim 3, wherein the determination unit determines to alter the firmness of the surface of the target in a case where a gesture corresponding to a pressure change pattern satisfies a condition.
6. The information processing apparatus according to claim 3, wherein the determination unit determines an alteration amount in the firmness of the surface of the target according to the change amount in the pressure.
7. The information processing apparatus according to claim 3, wherein the acquisition unit acquires the sensor information including the pressure information indicating a pressure to each of a plurality of regions on the surface of the target, the estimation unit calculates, based on the pressure information, the change amount in the pressure applied by the object to each of the plurality of regions on the surface of the target, and the determination unit determines a region in which the change amount in the pressure satisfies a condition, among the plurality of regions on the surface of the target, as a region to undergo firmness alteration.
8. The information processing apparatus according to claim 1, wherein the acquisition unit acquires the sensor information including image information imaged by an image sensor, and the estimation unit estimates the contact mode of the object with the surface of the target based on the image information.
9. The information processing apparatus according to claim 8, wherein the estimation unit calculates an intersection between a feature point of the object and the surface of the target by using a moving direction of the object based on the image information, and the determination unit determines to alter the firmness of the surface of the target in a case where a distance between the feature point of the object and the intersection is a threshold or more.
10. The information processing apparatus according to claim 9, wherein the determination unit determines an alteration amount in the firmness of the surface of the target according to a moving speed of the object.
11. The information processing apparatus according to claim 9, wherein the estimation unit estimates a context related to the object based on the image information, and the determination unit determines to alter the firmness of the surface of the target in a case where the context estimated by the estimation unit satisfies a condition.
12. The information processing apparatus according to claim 11, wherein the determination unit determines an alteration amount in the firmness of the surface of the target according to the context.

- 13.** The information processing apparatus according to claim 8, wherein the acquisition unit acquires the sensor information including the image information obtained by imaging a user by the image sensor, the estimation unit estimates information related to the user based on the image information, and the determination unit determines whether to alter the firmness of the surface of the target based on the information related to the user estimated by the estimation unit.
- 14.** The information processing apparatus according to claim 8, wherein the estimation unit estimates the contact mode of the object with respect to a plurality of regions on the surface of the target based on the image information, and the determination unit determines a region in which the contact mode satisfies a condition among the plurality of regions on the surface of the target, as a region to undergo firmness alteration.
- 15.** An information processing method comprising: acquiring sensor information detected by a sensor; estimating, based on the acquired sensor information, a contact mode of an object with respect to a target of which a firmness of a surface is alterable; and determining whether to alter the firmness of the surface of the target according to the estimated contact mode of the object.
- 16.** A firmness alterable module constituting at least a part of a surface of a target, the firmness alterable module comprising: a sheet-like coupling portion that is disposed along the surface and is flexible in a direction intersecting the surface; a block portion having a plurality of blocks disposed side by side along a back surface and coupled to the back surface, the back surface being a surface opposite to the surface of the coupling portion; and a softness/firmness controller that is disposed along one surface of each of the plurality of blocks with the block portion interposed between the softness/firmness controller and the coupling portion, the softness/firmness controller being capable of altering firmness.
- 17.** The firmness alterable module according to claim 16, wherein each of the plurality of blocks is a hexahedron in which a facing surface facing the one surface is coupled to the coupling portion, and surfaces other than the one surface and the facing surface are disposed adjacent to surfaces of other blocks.
- 18.** The firmness alterable module according to claim 16, wherein the softness/firmness controller includes: a flexible member disposed along one surface of each of the plurality of blocks; and a pressing mechanism that presses the flexible member toward the block portion from a position where the block portion and the flexible member are sandwiched.
- 19.** The firmness alterable module according to claim 18, wherein the pressing mechanism includes: a movable portion having a plate shape along the flexible member at a position sandwiching the block portion and the flexible member; and a drive unit that moves the movable portion in a direction intersecting the surface.
- 20.** The firmness alterable module according to claim 16, further comprising a cover portion that covers the coupling portion, the block portion, and the softness/firmness controller, and to which an object comes in contact with a surface being an opposite surface of a surface running along the coupling portion.
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